

Solve for x and y to get $x = 0, y = 0$. Hence, F is nonsingular and so invertible.

(b) To find a formula for F^{-1} , we set $F(x, y) = (s, t)$ and so $F^{-1}(s, t) = (x, y)$. We have

$$(2x + y, 3x + 2y) = (s, t) \quad \text{or} \quad \begin{aligned} 2x + y &= s \\ 3x + 2y &= t \end{aligned}$$

Solve for x and y in terms of s and t to obtain $x = 2s - t, y = -3s + 2t$. Thus,

$$F^{-1}(s, t) = (2s - t, -3s + 2t) \quad \text{or} \quad F^{-1}(x, y) = (2x - y, -3x + 2y)$$

where we rewrite the formula for F^{-1} using x and y instead of s and t .

SOLVED PROBLEMS

Mappings

5.1. State whether each diagram in Fig. 5-3 defines a mapping from $A = \{a, b, c\}$ into $B = \{x, y, z\}$.

- (a) No. There is nothing assigned to the element $b \in A$.
- (b) No. Two elements, x and z , are assigned to $c \in A$.
- (c) Yes.

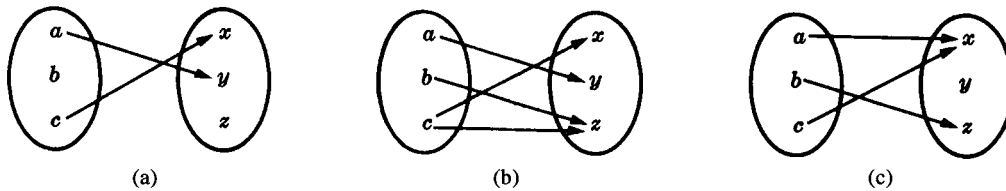


Figure 5-3

5.2. Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be defined by Fig. 5-4.

- (a) Find the composition mapping $(g \circ f): A \rightarrow C$.
- (b) Find the images of the mappings $f, g, g \circ f$.

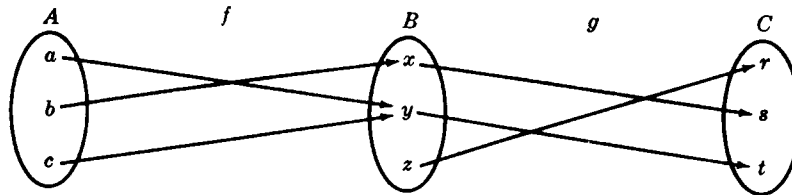


Figure 5-4

- (a) Use the definition of the composition mapping to compute

$$\begin{aligned} (g \circ f)(a) &= g(f(a)) = g(x) = r, & (g \circ f)(b) &= g(f(b)) = g(y) = s \\ (g \circ f)(c) &= g(f(c)) = g(z) = t \end{aligned}$$

Observe that we arrive at the same answer if we “follow the arrows” in Fig. 5-4:

$$a \rightarrow y \rightarrow t, \quad b \rightarrow x \rightarrow s, \quad c \rightarrow y \rightarrow t$$

- (b) By Fig. 5-4, the image values under the mapping f are x and y , and the image values under g are r, s, t .

Hence,

$$\text{Im } f = \{x, y\} \quad \text{and} \quad \text{Im } g = \{r, s, t\}$$

Also, by part (a), the image values under the composition mapping $g \circ f$ are t and s ; accordingly, $\text{Im } g \circ f = \{s, t\}$. Note that the images of g and $g \circ f$ are different.

- 5.3.** Consider the mapping $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by $F(x, y, z) = (yz, x^2)$. Find
 (a) $F(2, 3, 4)$; (b) $F(5, -2, 7)$; (c) $F^{-1}(0, 0)$, that is, all $v \in \mathbf{R}^3$ such that $F(v) = 0$.

- (a) Substitute in the formula for F to get $F(2, 3, 4) = (3 \cdot 4, 2^2) = (12, 4)$.
 (b) $F(5, -2, 7) = (-2 \cdot 7, 5^2) = (-14, 25)$.
 (c) Set $F(v) = 0$, where $v = (x, y, z)$, and then solve for x, y, z :

$$F(x, y, z) = (yz, x^2) = (0, 0) \quad \text{or} \quad yz = 0, x^2 = 0$$

Thus, $x = 0$ and either $y = 0$ or $z = 0$. In other words, $x = 0, y = 0$ or $x = 0, z = 0$ —that is, the z -axis and the y -axis.

- 5.4.** Consider the mapping $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $F(x, y) = (3y, 2x)$. Let S be the unit circle in $\mathbf{R}^2$, that is, the solution set of $x^2 + y^2 = 1$. (a) Describe $F(S)$. (b) Find $F^{-1}(S)$.

- (a) Let (a, b) be an element of $F(S)$. Then there exists $(x, y) \in S$ such that $F(x, y) = (a, b)$. Hence,

$$(3y, 2x) = (a, b) \quad \text{or} \quad 3y = a, 2x = b \quad \text{or} \quad y = \frac{a}{3}, x = \frac{b}{2}$$

Because $(x, y) \in S$ —that is, $x^2 + y^2 = 1$ —we have

$$\left(\frac{b}{2}\right)^2 + \left(\frac{a}{3}\right)^2 = 1 \quad \text{or} \quad \frac{a^2}{9} + \frac{b^2}{4} = 1$$

Thus, $F(S)$ is an ellipse.

- (b) Let $F(x, y) = (a, b)$, where $(a, b) \in S$. Then $(3y, 2x) = (a, b)$ or $3y = a, 2x = b$. Because $(a, b) \in S$, we have $a^2 + b^2 = 1$. Thus, $(3y)^2 + (2x)^2 = 1$. Accordingly, $F^{-1}(S)$ is the ellipse $4x^2 + 9y^2 = 1$.

- 5.5.** Let the mappings $f: A \rightarrow B, g: B \rightarrow C, h: C \rightarrow D$ be defined by Fig. 5-5. Determine whether or not each function is (a) one-to-one; (b) onto; (c) invertible (i.e., has an inverse).

- (a) The mapping $f: A \rightarrow B$ is one-to-one, as each element of A has a different image. The mapping $g: B \rightarrow C$ is not one-to-one, because x and z both have the same image 4. The mapping $h: C \rightarrow D$ is one-to-one.
 (b) The mapping $f: A \rightarrow B$ is not onto, because $w \in B$ is not the image of any element of A . The mapping $g: B \rightarrow C$ is onto, as each element of C is the image of some element of B . The mapping $h: C \rightarrow D$ is also onto.
 (c) A mapping has an inverse if and only if it is one-to-one and onto. Hence, only h has an inverse.

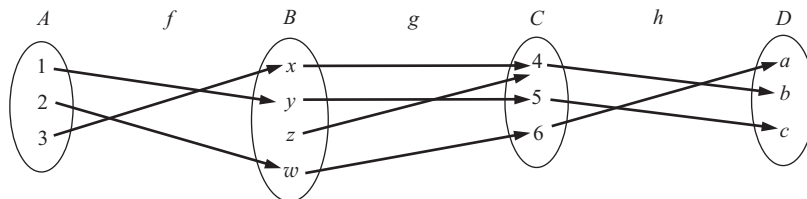


Figure 5-5

5.6. Suppose $f: A \rightarrow B$ and $g: B \rightarrow C$. Hence, $(g \circ f): A \rightarrow C$ exists. Prove

- (a) If f and g are one-to-one, then $g \circ f$ is one-to-one.
 - (b) If f and g are onto mappings, then $g \circ f$ is an onto mapping.
 - (c) If $g \circ f$ is one-to-one, then f is one-to-one.
 - (d) If $g \circ f$ is an onto mapping, then g is an onto mapping.
- (a) Suppose $(g \circ f)(x) = (g \circ f)(y)$. Then $g(f(x)) = g(f(y))$. Because g is one-to-one, $f(x) = f(y)$. Because f is one-to-one, $x = y$. We have proven that $(g \circ f)(x) = (g \circ f)(y)$ implies $x = y$; hence $g \circ f$ is one-to-one.
 - (b) Suppose $c \in C$. Because g is onto, there exists $b \in B$ for which $g(b) = c$. Because f is onto, there exists $a \in A$ for which $f(a) = b$. Thus, $(g \circ f)(a) = g(f(a)) = g(b) = c$. Hence, $g \circ f$ is onto.
 - (c) Suppose f is not one-to-one. Then there exist distinct elements $x, y \in A$ for which $f(x) = f(y)$. Thus, $(g \circ f)(x) = g(f(x)) = g(f(y)) = (g \circ f)(y)$. Hence, $g \circ f$ is not one-to-one. Therefore, if $g \circ f$ is one-to-one, then f must be one-to-one.
 - (d) If $a \in A$, then $(g \circ f)(a) = g(f(a)) \in g(B)$. Hence, $(g \circ f)(A) \subseteq g(B)$. Suppose g is not onto. Then $g(B)$ is properly contained in C and so $(g \circ f)(A)$ is properly contained in C ; thus, $g \circ f$ is not onto. Accordingly, if $g \circ f$ is onto, then g must be onto.

5.7. Prove that $f: A \rightarrow B$ has an inverse if and only if f is one-to-one and onto.

Suppose f has an inverse—that is, there exists a function $f^{-1}: B \rightarrow A$ for which $f^{-1} \circ f = \mathbf{1}_A$ and $f \circ f^{-1} = \mathbf{1}_B$. Because $\mathbf{1}_A$ is one-to-one, f is one-to-one by Problem 5.6(c), and because $\mathbf{1}_B$ is onto, f is onto by Problem 5.6(d); that is, f is both one-to-one and onto.

Now suppose f is both one-to-one and onto. Then each $b \in B$ is the image of a unique element in A , say b^* . Thus, if $f(a) = b$, then $a = b^*$; hence, $f(b^*) = b$. Now let g denote the mapping from B to A defined by $b \mapsto b^*$. We have

- (i) $(g \circ f)(a) = g(f(a)) = g(b) = b^* = a$ for every $a \in A$; hence, $g \circ f = \mathbf{1}_A$.
- (ii) $(f \circ g)(b) = f(g(b)) = f(b^*) = b$ for every $b \in B$; hence, $f \circ g = \mathbf{1}_B$.

Accordingly, f has an inverse. Its inverse is the mapping g .

5.8. Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be defined by $f(x) = 2x - 3$. Now f is one-to-one and onto; hence, f has an inverse mapping f^{-1} . Find a formula for f^{-1} .

Let y be the image of x under the mapping f ; that is, $y = f(x) = 2x - 3$. Hence, x will be the image of y under the inverse mapping f^{-1} . Thus, solve for x in terms of y in the above equation to obtain $x = \frac{1}{2}(y + 3)$. Then the formula defining the inverse function is $f^{-1}(y) = \frac{1}{2}(y + 3)$, or, using x instead of y , $f^{-1}(x) = \frac{1}{2}(x + 3)$.

Linear Mappings

5.9. Suppose the mapping $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ is defined by $F(x, y) = (x + y, x)$. Show that F is linear.

We need to show that $F(v + w) = F(v) + F(w)$ and $F(kv) = kF(v)$, where u and v are any elements of $\mathbf{R}^2$ and k is any scalar. Let $v = (a, b)$ and $w = (a', b')$. Then

$$v + w = (a + a', b + b') \quad \text{and} \quad kv = (ka, kb)$$

We have $F(v) = (a + b, a)$ and $F(w) = (a' + b', a')$. Thus,

$$\begin{aligned} F(v + w) &= F(a + a', b + b') = (a + a' + b + b', a + a') \\ &= (a + b, a) + (a' + b', a') = F(v) + F(w) \end{aligned}$$

and

$$F(kv) = F(ka, kb) = (ka + kb, ka) = k(a + b, a) = kF(v)$$

Because v, w, k were arbitrary, F is linear.

5.10. Suppose $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ is defined by $F(x, y, z) = (x + y + z, 2x - 3y + 4z)$. Show that F is linear.

We argue via matrices. Writing vectors as columns, the mapping F may be written in the form $F(v) = Av$, where $v = [x, y, z]^T$ and

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -3 & 4 \end{bmatrix}$$

Then, using properties of matrices, we have

$$F(v + w) = A(v + w) = Av + Aw = F(v) + F(w)$$

and

$$F(kv) = A(kv) = k(Av) = kF(v)$$

Thus, F is linear.

5.11. Show that the following mappings are not linear:

- (a) $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $F(x, y) = (xy, x)$
- (b) $F: \mathbf{R}^2 \rightarrow \mathbf{R}^3$ defined by $F(x, y) = (x + 3, 2y, x + y)$
- (c) $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by $F(x, y, z) = (|x|, y + z)$
- (a) Let $v = (1, 2)$ and $w = (3, 4)$; then $v + w = (4, 6)$. Also,

$$F(v) = (1(2), 1) = (2, 1) \quad \text{and} \quad F(w) = (3(4), 3) = (12, 3)$$

Hence,

$$F(v + w) = (4(6), 4) = (24, 6) \neq F(v) + F(w)$$

- (b) Because $F(0, 0) = (3, 0, 0) \neq (0, 0, 0)$, F cannot be linear.

- (c) Let $v = (1, 2, 3)$ and $k = -3$. Then $kv = (-3, -6, -9)$. We have

$$F(v) = (1, 5) \quad \text{and} \quad kF(v) = -3(1, 5) = (-3, -15).$$

Thus,

$$F(kv) = F(-3, -6, -9) = (3, -15) \neq kF(v)$$

Accordingly, F is not linear.

5.12. Let V be the vector space of n -square real matrices. Let M be an arbitrary but fixed matrix in V . Let $F: V \rightarrow V$ be defined by $F(A) = AM + MA$, where A is any matrix in V . Show that F is linear.

For any matrices A and B in V and any scalar k , we have

$$\begin{aligned} F(A + B) &= (A + B)M + M(A + B) = AM + BM + MA + MB \\ &= (AM + MA) + (BM + MB) = F(A) + F(B) \end{aligned}$$

and

$$F(kA) = (kA)M + M(kA) = k(AM) + k(MA) = k(AM + MA) = kF(A)$$

Thus, F is linear.

5.13. Prove Theorem 5.2: Let V and U be vector spaces over a field K . Let $\{v_1, v_2, \dots, v_n\}$ be a basis of V and let $u_1, u_2, \dots, u_n$ be any vectors in U . Then there exists a unique linear mapping $F: V \rightarrow U$ such that $F(v_1) = u_1, F(v_2) = u_2, \dots, F(v_n) = u_n$.

There are three steps to the proof of the theorem: (1) Define the mapping $F: V \rightarrow U$ such that $F(v_i) = u_i, i = 1, \dots, n$. (2) Show that F is linear. (3) Show that F is unique.

Step 1. Let $v \in V$. Because $\{v_1, \dots, v_n\}$ is a basis of V , there exist unique scalars $a_1, \dots, a_n \in K$ for which $v = a_1v_1 + a_2v_2 + \dots + a_nv_n$. We define $F: V \rightarrow U$ by

$$F(v) = a_1u_1 + a_2u_2 + \dots + a_nu_n$$

(Because the a_i are unique, the mapping F is well defined.) Now, for $i = 1, \dots, n$,

$$v_i = 0v_1 + \dots + 1v_i + \dots + 0v_n$$

Hence,

$$F(v_i) = 0u_1 + \dots + 1u_i + \dots + 0u_n = u_i$$

Thus, the first step of the proof is complete.

Step 2. Suppose $v = a_1v_1 + a_2v_2 + \dots + a_nv_n$ and $w = b_1v_1 + b_2v_2 + \dots + b_nv_n$. Then

$$v + w = (a_1 + b_1)v_1 + (a_2 + b_2)v_2 + \dots + (a_n + b_n)v_n$$

and, for any $k \in K$, $kv = ka_1v_1 + ka_2v_2 + \dots + ka_nv_n$. By definition of the mapping F ,

$$F(v) = a_1u_1 + a_2u_2 + \dots + a_nu_n \quad \text{and} \quad F(w) = b_1u_1 + b_2u_2 + \dots + b_nu_n$$

Hence,

$$\begin{aligned} F(v + w) &= (a_1 + b_1)u_1 + (a_2 + b_2)u_2 + \dots + (a_n + b_n)u_n \\ &= (a_1u_1 + a_2u_2 + \dots + a_nu_n) + (b_1u_1 + b_2u_2 + \dots + b_nu_n) \\ &= F(v) + F(w) \end{aligned}$$

and

$$F(kv) = k(a_1u_1 + a_2u_2 + \dots + a_nu_n) = kF(v)$$

Thus, F is linear.

Step 3. Suppose $G: V \rightarrow U$ is linear and $G(v_i) = u_i, i = 1, \dots, n$. Let

$$v = a_1v_1 + a_2v_2 + \dots + a_nv_n$$

Then

$$\begin{aligned} G(v) &= G(a_1v_1 + a_2v_2 + \dots + a_nv_n) = a_1G(v_1) + a_2G(v_2) + \dots + a_nG(v_n) \\ &= a_1u_1 + a_2u_2 + \dots + a_nu_n = F(v) \end{aligned}$$

Because $G(v) = F(v)$ for every $v \in V$, $G = F$. Thus, F is unique and the theorem is proved.

5.14. Let $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be the linear mapping for which $F(1, 2) = (2, 3)$ and $F(0, 1) = (1, 4)$. [Note that $\{(1, 2), (0, 1)\}$ is a basis of $\mathbf{R}^2$, so such a linear map F exists and is unique by Theorem 5.2.] Find a formula for F ; that is, find $F(a, b)$.

Write (a, b) as a linear combination of $(1, 2)$ and $(0, 1)$ using unknowns x and y ,

$$(a, b) = x(1, 2) + y(0, 1) = (x, 2x + y), \quad \text{so} \quad a = x, \quad b = 2x + y$$

Solve for x and y in terms of a and b to get $x = a, \quad y = -2a + b$. Then

$$F(a, b) = xF(1, 2) + yF(0, 1) = a(2, 3) + (-2a + b)(1, 4) = (b, -5a + 4b)$$

5.15. Suppose a linear mapping $F: V \rightarrow U$ is one-to-one and onto. Show that the inverse mapping $F^{-1}: U \rightarrow V$ is also linear.

Suppose $u, u' \in U$. Because F is one-to-one and onto, there exist unique vectors $v, v' \in V$ for which $F(v) = u$ and $F(v') = u'$. Because F is linear, we also have

$$F(v + v') = F(v) + F(v') = u + u' \quad \text{and} \quad F(kv) = kF(v) = ku$$

By definition of the inverse mapping,

$$F^{-1}(u) = v, \quad F^{-1}(u') = v', \quad F^{-1}(u + u') = v + v', \quad F^{-1}(ku) = kv.$$

Then

$$F^{-1}(u + u') = v + v' = F^{-1}(u) + F^{-1}(u') \quad \text{and} \quad F^{-1}(ku) = kv = kF^{-1}(u)$$

Thus, F^{-1} is linear.

Kernel and Image of Linear Mappings

5.16. Let $F: \mathbf{R}^4 \rightarrow \mathbf{R}^3$ be the linear mapping defined by

$$F(x, y, z, t) = (x - y + z + t, \quad x + 2z - t, \quad x + y + 3z - 3t)$$

Find a basis and the dimension of (a) the image of F , (b) the kernel of F .

(a) Find the images of the usual basis of $\mathbf{R}^4$:

$$F(1, 0, 0, 0) = (1, 1, 1), \quad F(0, 0, 1, 0) = (1, 2, 3)$$

$$F(0, 1, 0, 0) = (-1, 0, 1), \quad F(0, 0, 0, 1) = (1, -1, -3)$$

By Proposition 5.4, the image vectors span $\text{Im } F$. Hence, form the matrix whose rows are these image vectors, and row reduce to echelon form:

$$\begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 1 & 2 & 3 \\ 1 & -1 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & -2 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus, $(1, 1, 1)$ and $(0, 1, 2)$ form a basis for $\text{Im } F$; hence, $\dim(\text{Im } F) = 2$.

(b) Set $F(v) = 0$, where $v = (x, y, z, t)$; that is, set

$$F(x, y, z, t) = (x - y + z + t, \quad x + 2z - t, \quad x + y + 3z - 3t) = (0, 0, 0)$$

Set corresponding entries equal to each other to form the following homogeneous system whose solution space is $\text{Ker } F$:

$$\begin{array}{rcl} x - y + z + t = 0 & & x - y + z + t = 0 \\ x + 2z - t = 0 & \text{or} & y + z - 2t = 0 \\ x + y + 3z - 3t = 0 & & 2y + 2z - 4t = 0 \end{array} \quad \text{or} \quad \begin{array}{rcl} x - y + z + t = 0 & & \\ y + z - 2t = 0 & & \\ y + z - 2t = 0 & & \end{array}$$

The free variables are z and t . Hence, $\dim(\text{Ker } F) = 2$.

(i) Set $z = -1, t = 0$ to obtain the solution $(2, 1, -1, 0)$.

(ii) Set $z = 0, t = 1$ to obtain the solution $(1, 2, 0, 1)$.

Thus, $(2, 1, -1, 0)$ and $(1, 2, 0, 1)$ form a basis of $\text{Ker } F$.

[As expected, $\dim(\text{Im } F) + \dim(\text{Ker } F) = 2 + 2 = 4 = \dim \mathbf{R}^4$, the domain of F .]

5.17. Let $G: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be the linear mapping defined by

$$G(x, y, z) = (x + 2y - z, \quad y + z, \quad x + y - 2z)$$

Find a basis and the dimension of (a) the image of G , (b) the kernel of G .

(a) Find the images of the usual basis of $\mathbf{R}^3$:

$$G(1, 0, 0) = (1, 0, 1), \quad G(0, 1, 0) = (2, 1, 1), \quad G(0, 0, 1) = (-1, 1, -2)$$

By Proposition 5.4, the image vectors span $\text{Im } G$. Hence, form the matrix M whose rows are these image vectors, and row reduce to echelon form:

$$M = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ -1 & 1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus, $(1, 0, 1)$ and $(0, 1, -1)$ form a basis for $\text{Im } G$; hence, $\dim(\text{Im } G) = 2$.

(b) Set $G(v) = 0$, where $v = (x, y, z)$; that is,

$$G(x, y, z) = (x + 2y - z, \quad y + z, \quad x + y - 2z) = (0, 0, 0)$$

Set corresponding entries equal to each other to form the following homogeneous system whose solution space is $\text{Ker } G$:

$$\begin{array}{rcl} x + 2y - z = 0 & & x + 2y - z = 0 \\ y + z = 0 & \text{or} & y + z = 0 \\ x + y - 2z = 0 & & -y - z = 0 \end{array} \quad \text{or} \quad \begin{array}{rcl} x + 2y - z = 0 & & x + 2y - z = 0 \\ y + z = 0 & & y + z = 0 \end{array}$$

The only free variable is z ; hence, $\dim(\text{Ker } G) = 1$. Set $z = 1$; then $y = -1$ and $x = 3$. Thus, $(3, -1, 1)$ forms a basis of $\text{Ker } G$. [As expected, $\dim(\text{Im } G) + \dim(\text{Ker } G) = 2 + 1 = 3 = \dim \mathbf{R}^3$, the domain of G .]

5.18. Consider the matrix mapping $A: \mathbf{R}^4 \rightarrow \mathbf{R}^3$, where $A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 3 & 5 & -2 \\ 3 & 8 & 13 & -3 \end{bmatrix}$. Find a basis and the dimension of (a) the image of A , (b) the kernel of A .

(a) The column space of A is equal to $\text{Im } A$. Now reduce A^T to echelon form:

$$A^T = \begin{bmatrix} 1 & 1 & 3 \\ 2 & 3 & 8 \\ 3 & 5 & 13 \\ 1 & -2 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 2 \\ 0 & 2 & 4 \\ 0 & -3 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus, $\{(1, 1, 3), (0, 1, 2)\}$ is a basis of $\text{Im } A$, and $\dim(\text{Im } A) = 2$.

(b) Here $\text{Ker } A$ is the solution space of the homogeneous system $AX = 0$, where $X = \{x, y, z, t\}^T$. Thus, reduce the matrix A of coefficients to echelon form:

$$\begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 1 & 2 & -3 \\ 0 & 2 & 4 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 1 & 2 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{or} \quad \begin{array}{rcl} x + 2y + 3z + t & = & 0 \\ y + 2z - 3t & = & 0 \end{array}$$

The free variables are z and t . Thus, $\dim(\text{Ker } A) = 2$.

(i) Set $z = 1, t = 0$ to get the solution $(1, -2, 1, 0)$.

(ii) Set $z = 0, t = 1$ to get the solution $(-7, 3, 0, 1)$.

Thus, $(1, -2, 1, 0)$ and $(-7, 3, 0, 1)$ form a basis for $\text{Ker } A$.

5.19. Find a linear map $F: \mathbf{R}^3 \rightarrow \mathbf{R}^4$ whose image is spanned by $(1, 2, 0, -4)$ and $(2, 0, -1, -3)$.

Form a 4×3 matrix whose columns consist only of the given vectors, say

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 0 & 0 \\ 0 & -1 & -1 \\ -4 & -3 & -3 \end{bmatrix}$$

Recall that A determines a linear map $A: \mathbf{R}^3 \rightarrow \mathbf{R}^4$ whose image is spanned by the columns of A . Thus, A satisfies the required condition.

5.20. Suppose $f: V \rightarrow U$ is linear with kernel W , and that $f(v) = u$. Show that the “coset” $v + W = \{v + w: w \in W\}$ is the preimage of u ; that is, $f^{-1}(u) = v + W$.

We must prove that (i) $f^{-1}(u) \subseteq v + W$ and (ii) $v + W \subseteq f^{-1}(u)$.

We first prove (i). Suppose $v' \in f^{-1}(u)$. Then $f(v') = u$, and so

$$f(v' - v) = f(v') - f(v) = u - u = 0$$

that is, $v' - v \in W$. Thus, $v' = v + (v' - v) \in v + W$, and hence $f^{-1}(u) \subseteq v + W$.

Now we prove (ii). Suppose $v' \in v + W$. Then $v' = v + w$, where $w \in W$. Because W is the kernel of f , we have $f(w) = 0$. Accordingly,

$$f(v') = f(v + w) = f(v) + f(w) = f(v) + 0 = f(v) = u$$

Thus, $v' \in f^{-1}(u)$, and so $v + W \subseteq f^{-1}(u)$.

Both inclusions imply $f^{-1}(u) = v + W$.

5.21. Suppose $F: V \rightarrow U$ and $G: U \rightarrow W$ are linear. Prove

$$(a) \text{ rank}(G \circ F) \leq \text{rank}(G), \quad (b) \text{ rank}(G \circ F) \leq \text{rank}(F).$$

(a) Because $F(V) \subseteq U$, we also have $G(F(V)) \subseteq G(U)$, and so $\dim[G(F(V))] \leq \dim[G(U)]$. Then $\text{rank}(G \circ F) = \dim[(G \circ F)(V)] = \dim[G(F(V))] \leq \dim[G(U)] = \text{rank}(G)$.

(b) We have $\dim[G(F(V))] \leq \dim[F(V)]$. Hence,

$$\text{rank}(G \circ F) = \dim[(G \circ F)(V)] = \dim[G(F(V))] \leq \dim[F(V)] = \text{rank}(F)$$

5.22. Prove Theorem 5.3: Let $F: V \rightarrow U$ be linear. Then,

(a) $\text{Im } F$ is a subspace of U , (b) $\text{Ker } F$ is a subspace of V .

(a) Because $F(0) = 0$, we have $0 \in \text{Im } F$. Now suppose $u, u' \in \text{Im } F$ and $a, b \in K$. Because u and u' belong to the image of F , there exist vectors $v, v' \in V$ such that $F(v) = u$ and $F(v') = u'$. Then

$$F(av + bv') = aF(v) + bF(v') = au + bu' \in \text{Im } F$$

Thus, the image of F is a subspace of U .

(b) Because $F(0) = 0$, we have $0 \in \text{Ker } F$. Now suppose $v, w \in \text{Ker } F$ and $a, b \in K$. Because v and w belong to the kernel of F , $F(v) = 0$ and $F(w) = 0$. Thus,

$$F(av + bw) = aF(v) + bF(w) = a0 + b0 = 0 + 0 = 0, \quad \text{and so} \quad av + bw \in \text{Ker } F$$

Thus, the kernel of F is a subspace of V .

5.23. Prove Theorem 5.6: Suppose V has finite dimension and $F: V \rightarrow U$ is linear. Then

$$\dim V = \dim(\text{Ker } F) + \dim(\text{Im } F) = \text{nullity}(F) + \text{rank}(F)$$

Suppose $\dim(\text{Ker } F) = r$ and $\{w_1, \dots, w_r\}$ is a basis of $\text{Ker } F$, and suppose $\dim(\text{Im } F) = s$ and $\{u_1, \dots, u_s\}$ is a basis of $\text{Im } F$. (By Proposition 5.4, $\text{Im } F$ has finite dimension.) Because every $u_j \in \text{Im } F$, there exist vectors $v_1, \dots, v_s$ in V such that $F(v_1) = u_1, \dots, F(v_s) = u_s$. We claim that the set

$$B = \{w_1, \dots, w_r, v_1, \dots, v_s\}$$

is a basis of V ; that is, (i) B spans V , and (ii) B is linearly independent. Once we prove (i) and (ii), then $\dim V = r + s = \dim(\text{Ker } F) + \dim(\text{Im } F)$.

(i) B spans V . Let $v \in V$. Then $F(v) \in \text{Im } F$. Because the u_j span $\text{Im } F$, there exist scalars $a_1, \dots, a_s$ such that $F(v) = a_1u_1 + \dots + a_su_s$. Set $\hat{v} = a_1v_1 + \dots + a_sv_s - v$. Then

$$\begin{aligned} F(\hat{v}) &= F(a_1v_1 + \dots + a_sv_s - v) = a_1F(v_1) + \dots + a_sF(v_s) - F(v) \\ &= a_1u_1 + \dots + a_su_s - F(v) = 0 \end{aligned}$$

Thus, $\hat{v} \in \text{Ker } F$. Because the w_i span $\text{Ker } F$, there exist scalars $b_1, \dots, b_r$, such that

$$\hat{v} = b_1w_1 + \dots + b_rw_r = a_1v_1 + \dots + a_sv_s - v$$

Accordingly,

$$v = a_1v_1 + \dots + a_sv_s - b_1w_1 - \dots - b_rw_r$$

Thus, B spans V .

(ii) B is linearly independent. Suppose

$$x_1 w_1 + \cdots + x_r w_r + y_1 v_1 + \cdots + y_s v_s = 0 \quad (1)$$

where $x_i, y_j \in K$. Then

$$\begin{aligned} 0 &= F(0) = F(x_1 w_1 + \cdots + x_r w_r + y_1 v_1 + \cdots + y_s v_s) \\ &= x_1 F(w_1) + \cdots + x_r F(w_r) + y_1 F(v_1) + \cdots + y_s F(v_s) \end{aligned} \quad (2)$$

But $F(w_i) = 0$, since $w_i \in \text{Ker } F$, and $F(v_j) = u_j$. Substituting into (2), we will obtain $y_1 u_1 + \cdots + y_s u_s = 0$. Since the u_j are linearly independent, each $y_j = 0$. Substitution into (1) gives $x_1 w_1 + \cdots + x_r w_r = 0$. Since the w_i are linearly independent, each $x_i = 0$. Thus B is linearly independent.

Singular and Nonsingular Linear Maps, Isomorphisms

5.24. Determine whether or not each of the following linear maps is nonsingular. If not, find a nonzero vector v whose image is 0.

(a) $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $F(x, y) = (x - y, x - 2y)$.

(b) $G: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $G(x, y) = (2x - 4y, 3x - 6y)$.

(a) Find $\text{Ker } F$ by setting $F(v) = 0$, where $v = (x, y)$,

$$(x - y, x - 2y) = (0, 0) \quad \text{or} \quad \begin{array}{l} x - y = 0 \\ x - 2y = 0 \end{array} \quad \text{or} \quad \begin{array}{l} x - y = 0 \\ -y = 0 \end{array}$$

The only solution is $x = 0, y = 0$. Hence, F is nonsingular.

(b) Set $G(x, y) = (0, 0)$ to find $\text{Ker } G$:

$$(2x - 4y, 3x - 6y) = (0, 0) \quad \text{or} \quad \begin{array}{l} 2x - 4y = 0 \\ 3x - 6y = 0 \end{array} \quad \text{or} \quad \begin{array}{l} x - 2y = 0 \\ x - 2y = 0 \end{array}$$

The system has nonzero solutions, because y is a free variable. Hence, G is singular. Let $y = 1$ to obtain the solution $v = (2, 1)$, which is a nonzero vector, such that $G(v) = 0$.

5.25. The linear map $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $F(x, y) = (x - y, x - 2y)$ is nonsingular by the previous Problem 5.24. Find a formula for F^{-1} .

Set $F(x, y) = (a, b)$, so that $F^{-1}(a, b) = (x, y)$. We have

$$(x - y, x - 2y) = (a, b) \quad \text{or} \quad \begin{array}{l} x - y = a \\ x - 2y = b \end{array} \quad \text{or} \quad \begin{array}{l} x - y = a \\ y = a - b \end{array}$$

Solve for x and y in terms of a and b to get $x = 2a - b, y = a - b$. Thus,

$$F^{-1}(a, b) = (2a - b, a - b) \quad \text{or} \quad F^{-1}(x, y) = (2x - y, x - y)$$

(The second equation is obtained by replacing a and b by x and y , respectively.)

5.26. Let $G: \mathbf{R}^2 \rightarrow \mathbf{R}^3$ be defined by $G(x, y) = (x + y, x - 2y, 3x + y)$.

(a) Show that G is nonsingular. (b) Find a formula for G^{-1} .

(a) Set $G(x, y) = (0, 0, 0)$ to find $\text{Ker } G$. We have

$$(x + y, x - 2y, 3x + y) = (0, 0, 0) \quad \text{or} \quad \begin{array}{l} x + y = 0 \\ x - 2y = 0 \\ 3x + y = 0 \end{array}$$

The only solution is $x = 0, y = 0$; hence, G is nonsingular.

(b) Although G is nonsingular, it is not invertible, because $\mathbf{R}^2$ and $\mathbf{R}^3$ have different dimensions. (Thus, Theorem 5.9 does not apply.) Accordingly, G^{-1} does not exist.

- 5.27.** Suppose that $F: V \rightarrow U$ is linear and that V is of finite dimension. Show that V and the image of F have the same dimension if and only if F is nonsingular. Determine all nonsingular linear mappings $T: \mathbf{R}^4 \rightarrow \mathbf{R}^3$.

By Theorem 5.6, $\dim V = \dim(\operatorname{Im} F) + \dim(\operatorname{Ker} F)$. Hence, V and $\operatorname{Im} F$ have the same dimension if and only if $\dim(\operatorname{Ker} F) = 0$ or $\operatorname{Ker} F = \{0\}$ (i.e., if and only if F is nonsingular).

Because $\dim \mathbf{R}^3$ is less than $\dim \mathbf{R}^4$, we have that $\dim(\operatorname{Im} T)$ is less than the dimension of the domain $\mathbf{R}^4$ of T . Accordingly no linear mapping $T: \mathbf{R}^4 \rightarrow \mathbf{R}^3$ can be nonsingular.

- 5.28.** Prove Theorem 5.7: Let $F: V \rightarrow U$ be a nonsingular linear mapping. Then the image of any linearly independent set is linearly independent.

Suppose $v_1, v_2, \dots, v_n$ are linearly independent vectors in V . We claim that $F(v_1), F(v_2), \dots, F(v_n)$ are also linearly independent. Suppose $a_1 F(v_1) + a_2 F(v_2) + \dots + a_n F(v_n) = 0$, where $a_i \in K$. Because F is linear, $F(a_1 v_1 + a_2 v_2 + \dots + a_n v_n) = 0$. Hence,

$$a_1 v_1 + a_2 v_2 + \dots + a_n v_n \in \operatorname{Ker} F$$

But F is nonsingular—that is, $\operatorname{Ker} F = \{0\}$. Hence, $a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0$. Because the v_i are linearly independent, all the a_i are 0. Accordingly, the $F(v_i)$ are linearly independent. Thus, the theorem is proved.

- 5.29.** Prove Theorem 5.9: Suppose V has finite dimension and $\dim V = \dim U$. Suppose $F: V \rightarrow U$ is linear. Then F is an isomorphism if and only if F is nonsingular.

If F is an isomorphism, then only 0 maps to 0; hence, F is nonsingular. Conversely, suppose F is nonsingular. Then $\dim(\operatorname{Ker} F) = 0$. By Theorem 5.6, $\dim V = \dim(\operatorname{Ker} F) + \dim(\operatorname{Im} F)$. Thus,

$$\dim U = \dim V = \dim(\operatorname{Im} F)$$

Because U has finite dimension, $\operatorname{Im} F = U$. This means F maps V onto U . Thus, F is one-to-one and onto; that is, F is an isomorphism.

Operations with Linear Maps

- 5.30.** Define $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ and $G: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ by $F(x, y, z) = (2x, y + z)$ and $G(x, y, z) = (x - z, y)$. Find formulas defining the maps: (a) $F + G$, (b) $3F$, (c) $2F - 5G$.

$$(a) (F + G)(x, y, z) = F(x, y, z) + G(x, y, z) = (2x, y + z) + (x - z, y) = (3x - z, 2y + z)$$

$$(b) (3F)(x, y, z) = 3F(x, y, z) = 3(2x, y + z) = (6x, 3y + 3z)$$

$$(c) (2F - 5G)(x, y, z) = 2F(x, y, z) - 5G(x, y, z) = 2(2x, y + z) - 5(x - z, y) \\ = (4x, 2y + 2z) + (-5x + 5z, -5y) = (-x + 5z, -3y + 2z)$$

- 5.31.** Let $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ and $G: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be defined by $F(x, y, z) = (2x, y + z)$ and $G(x, y) = (y, x)$. Derive formulas defining the mappings: (a) $G \circ F$, (b) $F \circ G$.

$$(a) (G \circ F)(x, y, z) = G(F(x, y, z)) = G(2x, y + z) = (y + z, 2x)$$

(b) The mapping $F \circ G$ is not defined, because the image of G is not contained in the domain of F .

- 5.32.** Prove: (a) The zero mapping $\mathbf{0}$, defined by $\mathbf{0}(v) = 0 \in U$ for every $v \in V$, is the zero element of $\operatorname{Hom}(V, U)$. (b) The negative of $F \in \operatorname{Hom}(V, U)$ is the mapping $(-1)F$, that is, $-F = (-1)F$.

Let $F \in \operatorname{Hom}(V, U)$. Then, for every $v \in V$:

$$(a) (F + \mathbf{0})(v) = F(v) + \mathbf{0}(v) = F(v) + 0 = F(v)$$

Because $(F + \mathbf{0})(v) = F(v)$ for every $v \in V$, we have $F + \mathbf{0} = F$. Similarly, $\mathbf{0} + F = F$.

$$(b) (F + (-1)F)(v) = F(v) + (-1)F(v) = F(v) - F(v) = 0 = \mathbf{0}(v)$$

Thus, $F + (-1)F = \mathbf{0}$. Similarly $(-1)F + F = \mathbf{0}$. Hence, $-F = (-1)F$.

- 5.33.** Suppose $F_1, F_2, \dots, F_n$ are linear maps from V into U . Show that, for any scalars $a_1, a_2, \dots, a_n$, and for any $v \in V$,

$$(a_1F_1 + a_2F_2 + \dots + a_nF_n)(v) = a_1F_1(v) + a_2F_2(v) + \dots + a_nF_n(v)$$

The mapping a_1F_1 is defined by $(a_1F_1)(v) = a_1F(v)$. Hence, the theorem holds for $n = 1$. Accordingly, by induction,

$$\begin{aligned} (a_1F_1 + a_2F_2 + \dots + a_nF_n)(v) &= (a_1F_1)(v) + (a_2F_2 + \dots + a_nF_n)(v) \\ &= a_1F_1(v) + a_2F_2(v) + \dots + a_nF_n(v) \end{aligned}$$

- 5.34.** Consider linear mappings $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$, $G: \mathbf{R}^3 \rightarrow \mathbf{R}^2$, $H: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by

$$F(x, y, z) = (x + y + z, x + y), \quad G(x, y, z) = (2x + z, x + y), \quad H(x, y, z) = (2y, x)$$

Show that F, G, H are linearly independent [as elements of $\text{Hom}(\mathbf{R}^3, \mathbf{R}^2)$].

Suppose, for scalars $a, b, c \in K$,

$$aF + bG + cH = \mathbf{0} \tag{1}$$

(Here $\mathbf{0}$ is the zero mapping.) For $e_1 = (1, 0, 0) \in \mathbf{R}^3$, we have $\mathbf{0}(e_1) = (0, 0)$ and

$$\begin{aligned} (aF + bG + cH)(e_1) &= aF(1, 0, 0) + bG(1, 0, 0) + cH(1, 0, 0) \\ &= a(1, 1) + b(2, 1) + c(0, 1) = (a + 2b, a + b + c) \end{aligned}$$

Thus by (1), $(a + 2b, a + b + c) = (0, 0)$ and so

$$a + 2b = 0 \quad \text{and} \quad a + b + c = 0 \tag{2}$$

Similarly for $e_2 = (0, 1, 0) \in \mathbf{R}^3$, we have $\mathbf{0}(e_2) = (0, 0)$ and

$$\begin{aligned} (aF + bG + cH)(e_2) &= aF(0, 1, 0) + bG(0, 1, 0) + cH(0, 1, 0) \\ &= a(1, 1) + b(0, 1) + c(2, 0) = (a + 2c, a + b) \end{aligned}$$

Thus,

$$a + 2c = 0 \quad \text{and} \quad a + b = 0 \tag{3}$$

Using (2) and (3), we obtain

$$a = 0, \quad b = 0, \quad c = 0 \tag{4}$$

Because (1) implies (4), the mappings F, G, H are linearly independent.

- 5.35.** Let k be a nonzero scalar. Show that a linear map T is singular if and only if kT is singular. Hence, T is singular if and only if $-T$ is singular.

Suppose T is singular. Then $T(v) = 0$ for some vector $v \neq 0$. Hence,

$$(kT)(v) = kT(v) = k0 = 0$$

and so kT is singular.

Now suppose kT is singular. Then $(kT)(w) = 0$ for some vector $w \neq 0$. Hence,

$$T(kw) = kT(w) = (kT)(w) = 0$$

But $k \neq 0$ and $w \neq 0$ implies $kw \neq 0$. Thus, T is also singular.

- 5.36.** Find the dimension d of:

$$(a) \text{Hom}(\mathbf{R}^3, \mathbf{R}^4), \quad (b) \text{Hom}(\mathbf{R}^5, \mathbf{R}^3), \quad (c) \text{Hom}(\mathbf{P}_3(t), \mathbf{R}^2), \quad (d) \text{Hom}(\mathbf{M}_{2,3}, \mathbf{R}^4).$$

Use $\dim[\text{Hom}(V, U)] = mn$, where $\dim V = m$ and $\dim U = n$.

- (a) $d = 3(4) = 12$. (c) Because $\dim \mathbf{P}_3(t) = 4$, $d = 4(2) = 8$.
 (b) $d = 5(3) = 15$. (d) Because $\dim \mathbf{M}_{2,3} = 6$, $d = 6(4) = 24$.

5.37. Prove Theorem 5.11. Suppose $\dim V = m$ and $\dim U = n$. Then $\dim[\text{Hom}(V, U)] = mn$.

Suppose $\{v_1, \dots, v_m\}$ is a basis of V and $\{u_1, \dots, u_n\}$ is a basis of U . By Theorem 5.2, a linear mapping in $\text{Hom}(V, U)$ is uniquely determined by arbitrarily assigning elements of U to the basis elements v_i of V . We define

$$F_{ij} \in \text{Hom}(V, U), \quad i = 1, \dots, m, \quad j = 1, \dots, n$$

to be the linear mapping for which $F_{ij}(v_i) = u_j$, and $F_{ij}(v_k) = 0$ for $k \neq i$. That is, F_{ij} maps v_i into u_j and the other v 's into 0. Observe that $\{F_{ij}\}$ contains exactly mn elements; hence, the theorem is proved if we show that it is a basis of $\text{Hom}(V, U)$.

Proof that $\{F_{ij}\}$ generates $\text{Hom}(V, U)$. Consider an arbitrary function $F \in \text{Hom}(V, U)$. Suppose $F(v_1) = w_1, F(v_2) = w_2, \dots, F(v_m) = w_m$. Because $w_k \in U$, it is a linear combination of the u 's; say,

$$w_k = a_{k1}u_1 + a_{k2}u_2 + \dots + a_{kn}u_n, \quad k = 1, \dots, m, \quad a_{kj} \in K \quad (1)$$

Consider the linear mapping $G = \sum_{i=1}^m \sum_{j=1}^n a_{ij}F_{ij}$. Because G is a linear combination of the F_{ij} , the proof that $\{F_{ij}\}$ generates $\text{Hom}(V, U)$ is complete if we show that $F = G$.

We now compute $G(v_k), k = 1, \dots, m$. Because $F_{ij}(v_k) = 0$ for $k \neq i$ and $F_{ki}(v_k) = u_i$,

$$\begin{aligned} G(v_k) &= \sum_{i=1}^m \sum_{j=1}^n a_{ij}F_{ij}(v_k) = \sum_{j=1}^n a_{kj}F_{kj}(v_k) = \sum_{j=1}^n a_{kj}u_j \\ &= a_{k1}u_1 + a_{k2}u_2 + \dots + a_{kn}u_n \end{aligned}$$

Thus, by (1), $G(v_k) = w_k$ for each k . But $F(v_k) = w_k$ for each k . Accordingly, by Theorem 5.2, $F = G$; hence, $\{F_{ij}\}$ generates $\text{Hom}(V, U)$.

Proof that $\{F_{ij}\}$ is linearly independent. Suppose, for scalars $c_{ij} \in K$,

$$\sum_{i=1}^m \sum_{j=1}^n c_{ij}F_{ij} = \mathbf{0}$$

For $v_k, k = 1, \dots, m$,

$$\begin{aligned} 0 &= \mathbf{0}(v_k) = \sum_{i=1}^m \sum_{j=1}^n c_{ij}F_{ij}(v_k) = \sum_{j=1}^n c_{kj}F_{kj}(v_k) = \sum_{j=1}^n c_{kj}u_j \\ &= c_{k1}u_1 + c_{k2}u_2 + \dots + c_{kn}u_n \end{aligned}$$

But the u_i are linearly independent; hence, for $k = 1, \dots, m$, we have $c_{k1} = 0, c_{k2} = 0, \dots, c_{kn} = 0$. In other words, all the $c_{ij} = 0$, and so $\{F_{ij}\}$ is linearly independent.

5.38. Prove Theorem 5.12: (i) $G \circ (F + F') = G \circ F + G \circ F'$. (ii) $(G + G') \circ F = G \circ F + G' \circ F$. (iii) $k(G \circ F) = (kG) \circ F = G \circ (kF)$.

(i) For every $v \in V$,

$$\begin{aligned} (G \circ (F + F'))(v) &= G((F + F')(v)) = G(F(v) + F'(v)) \\ &= G(F(v)) + G(F'(v)) = (G \circ F)(v) + (G \circ F')(v) = (G \circ F + G \circ F')(v) \end{aligned}$$

Thus, $G \circ (F + F') = G \circ F + G \circ F'$.

(ii) For every $v \in V$,

$$\begin{aligned} ((G + G') \circ F)(v) &= (G + G')(F(v)) = G(F(v)) + G'(F(v)) \\ &= (G \circ F)(v) + (G' \circ F)(v) = (G \circ F + G' \circ F)(v) \end{aligned}$$

Thus, $(G + G') \circ F = G \circ F + G' \circ F$.

(iii) For every $v \in V$,

$$(k(G \circ F))(v) = k(G \circ F)(v) = k(G(F(v))) = (kG)(F(v)) = (kG \circ F)(v)$$

and

$$(k(G \circ F))(v) = k(G \circ F)(v) = k(G(F(v))) = G(kF(v)) = G((kF)(v)) = (G \circ kF)(v)$$

Accordingly, $k(G \circ F) = (kG) \circ F = G \circ (kF)$. (We emphasize that two mappings are shown to be equal by showing that each of them assigns the same image to each point in the domain.)

Algebra of Linear Maps

5.39. Let F and G be the linear operators on $\mathbf{R}^2$ defined by $F(x, y) = (y, x)$ and $G(x, y) = (0, x)$. Find formulas defining the following operators:

(a) $F + G$, (b) $2F - 3G$, (c) FG , (d) GF , (e) F^2 , (f) G^2 .

- (a) $(F + G)(x, y) = F(x, y) + G(x, y) = (y, x) + (0, x) = (y, 2x)$.
 (b) $(2F - 3G)(x, y) = 2F(x, y) - 3G(x, y) = 2(y, x) - 3(0, x) = (2y, -x)$.
 (c) $(FG)(x, y) = F(G(x, y)) = F(0, x) = (x, 0)$.
 (d) $(GF)(x, y) = G(F(x, y)) = G(y, x) = (0, y)$.
 (e) $F^2(x, y) = F(F(x, y)) = F(y, x) = (x, y)$. (Note that $F^2 = I$, the identity mapping.)
 (f) $G^2(x, y) = G(G(x, y)) = G(0, x) = (0, 0)$. (Note that $G^2 = \mathbf{0}$, the zero mapping.)

5.40. Consider the linear operator T on $\mathbf{R}^3$ defined by $T(x, y, z) = (2x, 4x - y, 2x + 3y - z)$.

(a) Show that T is invertible. Find formulas for (b) T^{-1} , (c) T^2 , (d) T^{-2} .

(a) Let $W = \text{Ker } T$. We need only show that T is nonsingular (i.e., that $W = \{0\}$). Set $T(x, y, z) = (0, 0, 0)$, which yields

$$T(x, y, z) = (2x, 4x - y, 2x + 3y - z) = (0, 0, 0)$$

Thus, W is the solution space of the homogeneous system

$$2x = 0, \quad 4x - y = 0, \quad 2x + 3y - z = 0$$

which has only the trivial solution $(0, 0, 0)$. Thus, $W = \{0\}$. Hence, T is nonsingular, and so T is invertible.

(b) Set $T(x, y, z) = (r, s, t)$ [and so $T^{-1}(r, s, t) = (x, y, z)$]. We have

$$(2x, 4x - y, 2x + 3y - z) = (r, s, t) \quad \text{or} \quad 2x = r, \quad 4x - y = s, \quad 2x + 3y - z = t$$

Solve for x, y, z in terms of r, s, t to get $x = \frac{1}{2}r$, $y = 2r - s$, $z = 7r - 3s - t$. Thus,

$$T^{-1}(r, s, t) = (\tfrac{1}{2}r, 2r - s, 7r - 3s - t) \quad \text{or} \quad T^{-1}(x, y, z) = (\tfrac{1}{2}x, 2x - y, 7x - 3y - z)$$

(c) Apply T twice to get

$$\begin{aligned} T^2(x, y, z) &= T(2x, 4x - y, 2x + 3y - z) \\ &= [4x, 4(2x) - (4x - y), 2(2x) + 3(4x - y) - (2x + 3y - z)] \\ &= (4x, 4x + y, 14x - 6y + z) \end{aligned}$$

(d) Apply T^{-1} twice to get

$$\begin{aligned} T^{-2}(x, y, z) &= T^{-2}(\tfrac{1}{2}x, 2x - y, 7x - 3y - z) \\ &= [\tfrac{1}{4}x, 2(\tfrac{1}{2}x) - (2x - y), 7(\tfrac{1}{2}x) - 3(2x - y) - (7x - 3y - z)] \\ &= (\tfrac{1}{4}x, -x + y, -\tfrac{19}{2}x + 6y + z) \end{aligned}$$

5.41. Let V be of finite dimension and let T be a linear operator on V for which $TR = I$, for some operator R on V . (We call R a *right inverse* of T .)

(a) Show that T is invertible. (b) Show that $R = T^{-1}$.

(c) Give an example showing that the above need not hold if V is of infinite dimension.

(a) Let $\dim V = n$. By Theorem 5.14, T is invertible if and only if T is onto; hence, T is invertible if and only if $\text{rank}(T) = n$. We have $n = \text{rank}(I) = \text{rank}(TR) \leq \text{rank}(T) \leq n$. Hence, $\text{rank}(T) = n$ and T is invertible.

(b) $TT^{-1} = T^{-1}T = I$. Then $R = IR = (T^{-1}T)R = T^{-1}(TR) = T^{-1}I = T^{-1}$.

(c) Let V be the space of polynomials in t over K ; say, $p(t) = a_0 + a_1t + a_2t^2 + \cdots + a_st^s$. Let T and R be the operators on V defined by

$$T(p(t)) = 0 + a_1 + a_2t + \cdots + a_st^{s-1} \quad \text{and} \quad R(p(t)) = a_0t + a_1t^2 + \cdots + a_st^{s+1}$$

We have

$$(TR)(p(t)) = T(R(p(t))) = T(a_0t + a_1t^2 + \cdots + a_st^{s+1}) = a_0 + a_1t + \cdots + a_st^s = p(t)$$

and so $TR = I$, the identity mapping. On the other hand, if $k \in K$ and $k \neq 0$, then

$$(RT)(k) = R(T(k)) = R(0) = 0 \neq k$$

Accordingly, $RT \neq I$.

5.42. Let F and G be linear operators on $\mathbf{R}^2$ defined by $F(x, y) = (0, x)$ and $G(x, y) = (x, 0)$. Show that

(a) $GF = \mathbf{0}$, the zero mapping, but $FG \neq \mathbf{0}$. (b) $G^2 = G$.

(a) $(GF)(x, y) = G(F(x, y)) = G(0, x) = (0, 0)$. Because GF assigns $\mathbf{0} = (0, 0)$ to every vector (x, y) in $\mathbf{R}^2$, it is the zero mapping; that is, $GF = \mathbf{0}$.

On the other hand, $(FG)(x, y) = F(G(x, y)) = F(x, 0) = (0, x)$. For example, $(FG)(2, 3) = (0, 2)$. Thus, $FG \neq \mathbf{0}$, as it does not assign $\mathbf{0} = (0, 0)$ to every vector in $\mathbf{R}^2$.

(b) For any vector (x, y) in $\mathbf{R}^2$, we have $G^2(x, y) = G(G(x, y)) = G(x, 0) = (x, 0) = G(x, y)$. Hence, $G^2 = G$.

5.43. Find the dimension of (a) $A(\mathbf{R}^4)$, (b) $A(\mathbf{P}_2(t))$, (c) $A(\mathbf{M}_{2,3})$.

Use $\dim[A(V)] = n^2$ where $\dim V = n$. Hence, (a) $\dim[A(\mathbf{R}^4)] = 4^2 = 16$, (b) $\dim[A(\mathbf{P}_2(t))] = 3^2 = 9$, (c) $\dim[A(\mathbf{M}_{2,3})] = 6^2 = 36$.

5.44. Let E be a linear operator on V for which $E^2 = E$. (Such an operator is called a *projection*.) Let U be the image of E , and let W be the kernel. Prove

(a) If $u \in U$, then $E(u) = u$ (i.e., E is the identity mapping on U).

(b) If $E \neq I$, then E is singular—that is, $E(v) = 0$ for some $v \neq 0$.

(c) $V = U \oplus W$.

(a) If $u \in U$, the image of E , then $E(v) = u$ for some $v \in V$. Hence, using $E^2 = E$, we have

$$u = E(v) = E^2(v) = E(E(v)) = E(u)$$

(b) If $E \neq I$, then for some $v \in V$, $E(v) = u$, where $v \neq u$. By (i), $E(u) = u$. Thus,

$$E(v - u) = E(v) - E(u) = u - u = 0, \quad \text{where} \quad v - u \neq 0$$

(c) We first show that $V = U + W$. Let $v \in V$. Set $u = E(v)$ and $w = v - E(v)$. Then

$$v = E(v) + v - E(v) = u + w$$

By definition, $u = E(v) \in U$, the image of E . We now show that $w \in W$, the kernel of E ,

$$E(w) = E(v - E(v)) = E(v) - E^2(v) = E(v) - E(v) = 0$$

and thus $w \in W$. Hence, $V = U + W$.

We next show that $U \cap W = \{0\}$. Let $v \in U \cap W$. Because $v \in U$, $E(v) = v$ by part (a). Because $v \in W$, $E(v) = 0$. Thus, $v = E(v) = 0$ and so $U \cap W = \{0\}$.

The above two properties imply that $V = U \oplus W$.

SUPPLEMENTARY PROBLEMS

Mappings

- 5.45.** Determine the number of different mappings from (a) $\{1, 2\}$ into $\{1, 2, 3\}$, (b) $\{1, 2, \dots, r\}$ into $\{1, 2, \dots, s\}$.
- 5.46.** Let $f: \mathbf{R} \rightarrow \mathbf{R}$ and $g: \mathbf{R} \rightarrow \mathbf{R}$ be defined by $f(x) = x^2 + 3x + 1$ and $g(x) = 2x - 3$. Find formulas defining the composition mappings: (a) $f \circ g$; (b) $g \circ f$; (c) $g \circ g$; (d) $f \circ f$.
- 5.47.** For each mappings $f: \mathbf{R} \rightarrow \mathbf{R}$ find a formula for its inverse: (a) $f(x) = 3x - 7$, (b) $f(x) = x^3 + 2$.
- 5.48.** For any mapping $f: A \rightarrow B$, show that $\mathbf{1}_B \circ f = f = f \circ \mathbf{1}_A$.

Linear Mappings

- 5.49.** Show that the following mappings are linear:
- (a) $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by $F(x, y, z) = (x + 2y - 3z, 4x - 5y + 6z)$.
- (b) $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $F(x, y) = (ax + by, cx + dy)$, where a, b, c, d belong to $\mathbf{R}$.
- 5.50.** Show that the following mappings are not linear:
- (a) $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $F(x, y) = (x^2, y^2)$.
- (b) $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by $F(x, y, z) = (x + 1, y + z)$.
- (c) $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $F(x, y) = (xy, y)$.
- (d) $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by $F(x, y, z) = (|x|, y + z)$.
- 5.51.** Find $F(a, b)$, where the linear map $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ is defined by $F(1, 2) = (3, -1)$ and $F(0, 1) = (2, 1)$.
- 5.52.** Find a 2×2 matrix A that maps
- (a) $(1, 3)^T$ and $(1, 4)^T$ into $(-2, 5)^T$ and $(3, -1)^T$, respectively.
- (b) $(2, -4)^T$ and $(-1, 2)^T$ into $(1, 1)^T$ and $(1, 3)^T$, respectively.
- 5.53.** Find a 2×2 singular matrix B that maps $(1, 1)^T$ into $(1, 3)^T$.
- 5.54.** Let V be the vector space of real n -square matrices, and let M be a fixed nonzero matrix in V . Show that the first two of the following mappings $T: V \rightarrow V$ are linear, but the third is not:
- (a) $T(A) = MA$, (b) $T(A) = AM + MA$, (c) $T(A) = M + A$.
- 5.55.** Give an example of a nonlinear map $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ such that $F^{-1}(0) = \{0\}$ but F is not one-to-one.
- 5.56.** Let $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be defined by $F(x, y) = (3x + 5y, 2x + 3y)$, and let S be the unit circle in $\mathbf{R}^2$. (S consists of all points satisfying $x^2 + y^2 = 1$.) Find (a) the image $F(S)$, (b) the preimage $F^{-1}(S)$.
- 5.57.** Consider the linear map $G: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ defined by $G(x, y, z) = (x + y + z, y - 2z, y - 3z)$ and the unit sphere S_2 in $\mathbf{R}^3$, which consists of the points satisfying $x^2 + y^2 + z^2 = 1$. Find (a) $G(S_2)$, (b) $G^{-1}(S_2)$.
- 5.58.** Let H be the plane $x + 2y - 3z = 4$ in $\mathbf{R}^3$ and let G be the linear map in Problem 5.57. Find (a) $G(H)$, (b) $G^{-1}(H)$.
- 5.59.** Let W be a subspace of V . The inclusion map, denoted by $i: W \hookrightarrow V$, is defined by $i(w) = w$ for every $w \in W$. Show that the inclusion map is linear.
- 5.60.** Suppose $F: V \rightarrow U$ is linear. Show that $F(-v) = -F(v)$.

Kernel and Image of Linear Mappings

- 5.61.** For each linear map F find a basis and the dimension of the kernel and the image of F :
- (a) $F: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ defined by $F(x, y, z) = (x + 2y - 3z, 2x + 5y - 4z, x + 4y + z)$,
- (b) $F: \mathbf{R}^4 \rightarrow \mathbf{R}^3$ defined by $F(x, y, z, t) = (x + 2y + 3z + 2t, 2x + 4y + 7z + 5t, x + 2y + 6z + 5t)$.

5.62. For each linear map G , find a basis and the dimension of the kernel and the image of G :

- (a) $G: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by $G(x, y, z) = (x + y + z, 2x + 2y + 2z)$,
 (b) $G: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by $G(x, y, z) = (x + y, y + z)$,
 (c) $G: \mathbf{R}^5 \rightarrow \mathbf{R}^3$ defined by

$$G(x, y, z, s, t) = (x + 2y + 2z + s + t, x + 2y + 3z + 2s - t, 3x + 6y + 8z + 5s - t).$$

5.63. Each of the following matrices determines a linear map from $\mathbf{R}^4$ into $\mathbf{R}^3$:

$$(a) A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & -1 & 2 & -1 \\ 1 & -3 & 2 & -2 \end{bmatrix}, (b) B = \begin{bmatrix} 1 & 0 & 2 & -1 \\ 2 & 3 & -1 & 1 \\ -2 & 0 & -5 & 3 \end{bmatrix}.$$

Find a basis as well as the dimension of the kernel and the image of each linear map.

5.64. Find a linear mapping $F: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ whose image is spanned by $(1, 2, 3)$ and $(4, 5, 6)$.

5.65. Find a linear mapping $G: \mathbf{R}^4 \rightarrow \mathbf{R}^3$ whose kernel is spanned by $(1, 2, 3, 4)$ and $(0, 1, 1, 1)$.

5.66. Let $V = \mathbf{P}_{10}(t)$, the vector space of polynomials of degree ≤ 10 . Consider the linear map $\mathbf{D}^4: V \rightarrow V$, where $\mathbf{D}^4$ denotes the fourth derivative $d^4(f)/dt^4$. Find a basis and the dimension of

- (a) the image of $\mathbf{D}^4$; (b) the kernel of $\mathbf{D}^4$.

5.67. Suppose $F: V \rightarrow U$ is linear. Show that (a) the image of any subspace of V is a subspace of U ;
 (b) the preimage of any subspace of U is a subspace of V .

5.68. Show that if $F: V \rightarrow U$ is onto, then $\dim U \leq \dim V$. Determine all linear maps $F: \mathbf{R}^3 \rightarrow \mathbf{R}^4$ that are onto.

5.69. Consider the zero mapping $\mathbf{0}: V \rightarrow U$ defined by $\mathbf{0}(v) = 0, \forall v \in V$. Find the kernel and the image of $\mathbf{0}$.

Operations with linear Mappings

5.70. Let $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ and $G: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ be defined by $F(x, y, z) = (y, x + z)$ and $G(x, y, z) = (2z, x - y)$. Find formulas defining the mappings $F + G$ and $3F - 2G$.

5.71. Let $H: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be defined by $H(x, y) = (y, 2x)$. Using the maps F and G in Problem 5.70, find formulas defining the mappings: (a) $H \circ F$ and $H \circ G$, (b) $F \circ H$ and $G \circ H$, (c) $H \circ (F + G)$ and $H \circ F + H \circ G$.

5.72. Show that the following mappings F, G, H are linearly independent:

- (a) $F, G, H \in \text{Hom}(\mathbf{R}^2, \mathbf{R}^2)$ defined by $F(x, y) = (x, 2y)$, $G(x, y) = (y, x + y)$, $H(x, y) = (0, x)$,
 (b) $F, G, H \in \text{Hom}(\mathbf{R}^3, \mathbf{R})$ defined by $F(x, y, z) = x + y + z$, $G(x, y, z) = y + z$, $H(x, y, z) = x - z$.

5.73. For $F, G \in \text{Hom}(V, U)$, show that $\text{rank}(F + G) \leq \text{rank}(F) + \text{rank}(G)$. (Here V has finite dimension.)

5.74. Let $F: V \rightarrow U$ and $G: U \rightarrow V$ be linear. Show that if F and G are nonsingular, then $G \circ F$ is nonsingular. Give an example where $G \circ F$ is nonsingular but G is not. [Hint: Let $\dim V < \dim U$.]

5.75. Find the dimension d of (a) $\text{Hom}(\mathbf{R}^2, \mathbf{R}^8)$, (b) $\text{Hom}(\mathbf{P}_4(t), \mathbf{R}^3)$, (c) $\text{Hom}(\mathbf{M}_{2,4}, \mathbf{P}_2(t))$.

5.76. Determine whether or not each of the following linear maps is nonsingular. If not, find a nonzero vector v whose image is 0; otherwise find a formula for the inverse map:

- (a) $F: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ defined by $F(x, y, z) = (x + y + z, 2x + 3y + 5z, x + 3y + 7z)$,
 (b) $G: \mathbf{R}^3 \rightarrow \mathbf{P}_2(t)$ defined by $G(x, y, z) = (x + y)t^2 + (x + 2y + 2z)t + y + z$,
 (c) $H: \mathbf{R}^2 \rightarrow \mathbf{P}_2(t)$ defined by $H(x, y) = (x + 2y)t^2 + (x - y)t + x + y$.

5.77. When can $\dim [\text{Hom}(V, U)] = \dim V$?

Algebra of Linear Operators

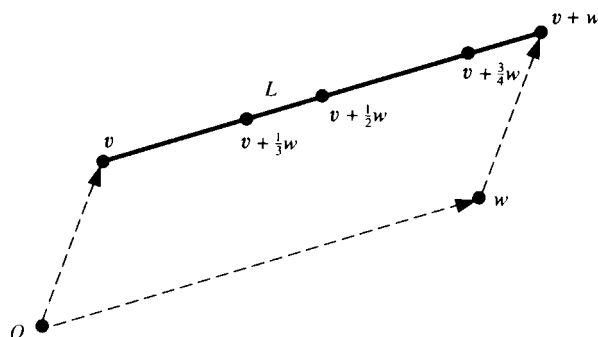
- 5.78.** Let F and G be the linear operators on $\mathbf{R}^2$ defined by $F(x, y) = (x + y, 0)$ and $G(x, y) = (-y, x)$. Find formulas defining the linear operators: (a) $F + G$, (b) $5F - 3G$, (c) FG , (d) GF , (e) F^2 , (f) G^2 .
- 5.79.** Show that each linear operator T on $\mathbf{R}^2$ is nonsingular and find a formula for T^{-1} , where (a) $T(x, y) = (x + 2y, 2x + 3y)$, (b) $T(x, y) = (2x - 3y, 3x - 4y)$.
- 5.80.** Show that each of the following linear operators T on $\mathbf{R}^3$ is nonsingular and find a formula for T^{-1} , where (a) $T(x, y, z) = (x - 3y - 2z, y - 4z, z)$; (b) $T(x, y, z) = (x + z, x - y, y)$.
- 5.81.** Find the dimension of $A(V)$, where (a) $V = \mathbf{R}^7$, (b) $V = \mathbf{P}_5(t)$, (c) $V = \mathbf{M}_{3,4}$.
- 5.82.** Which of the following integers can be the dimension of an algebra $A(V)$ of linear maps: 5, 9, 12, 25, 28, 36, 45, 64, 88, 100?
- 5.83.** Let T be the linear operator on $\mathbf{R}^2$ defined by $T(x, y) = (x + 2y, 3x + 4y)$. Find a formula for $f(T)$, where (a) $f(t) = t^2 + 2t - 3$, (b) $f(t) = t^2 - 5t - 2$.

Miscellaneous Problems

- 5.84.** Suppose $F: V \rightarrow U$ is linear and k is a nonzero scalar. Prove that the maps F and kF have the same kernel and the same image.
- 5.85.** Suppose F and G are linear operators on V and that F is nonsingular. Assume that V has finite dimension. Show that $\text{rank}(FG) = \text{rank}(GF) = \text{rank}(G)$.
- 5.86.** Suppose V has finite dimension. Suppose T is a linear operator on V such that $\text{rank}(T^2) = \text{rank}(T)$. Show that $\text{Ker } T \cap \text{Im } T = \{0\}$.
- 5.87.** Suppose $V = U \oplus W$. Let E_1 and E_2 be the linear operators on V defined by $E_1(v) = u$, $E_2(v) = w$, where $v = u + w$, $u \in U$, $w \in W$. Show that (a) $E_1^2 = E_1$ and $E_2^2 = E_2$ (i.e., that E_1 and E_2 are projections); (b) $E_1 + E_2 = I$, the identity mapping; (c) $E_1E_2 = 0$ and $E_2E_1 = 0$.
- 5.88.** Let E_1 and E_2 be linear operators on V satisfying parts (a), (b), (c) of Problem 5.88. Prove

$$V = \text{Im } E_1 \oplus \text{Im } E_2$$

- 5.89.** Let v and w be elements of a real vector space V . The line segment L from v to $v + w$ is defined to be the set of vectors $v + tw$ for $0 \leq t \leq 1$. (See Fig. 5.6.)
- (a) Show that the line segment L between vectors v and u consists of the points:
(i) $(1 - t)v + tu$ for $0 \leq t \leq 1$, (ii) $t_1v + t_2u$ for $t_1 + t_2 = 1$, $t_1 \geq 0$, $t_2 \geq 0$.
- (b) Let $F: V \rightarrow U$ be linear. Show that the image $F(L)$ of a line segment L in V is a line segment in U .

**Figure 5-6**

- 5.90.** Let $F: V \rightarrow U$ be linear and let W be a subspace of V . The *restriction* of F to W is the map $F|_W: W \rightarrow U$ defined by $F|_W(v) = F(v)$ for every v in W . Prove the following:
 (a) $F|_W$ is linear; (b) $\text{Ker}(F|_W) = (\text{Ker } F) \cap W$; (c) $\text{Im}(F|_W) = F(W)$.
- 5.91.** A subset X of a vector space V is said to be *convex* if the line segment L between any two points (vectors) $P, Q \in X$ is contained in X . (a) Show that the intersection of convex sets is convex; (b) suppose $F: V \rightarrow U$ is linear and X is convex. Show that $F(X)$ is convex.

ANSWERS TO SUPPLEMENTARY PROBLEMS

- 5.45.** (a) $3^2 = 9$, (b) s^r
- 5.46.** (a) $(f \circ g)(x) = 4x^2 + 1$, (b) $(g \circ f)(x) = 2x^2 + 6x - 1$, (c) $(g \circ g)(x) = 4x - 9$,
 (d) $(f \circ f)(x) = x^4 + 6x^3 + 14x^2 + 15x + 5$
- 5.47.** (a) $f^{-1}(x) = \frac{1}{3}(x + 7)$, (b) $f^{-1}(x) = \sqrt[3]{x - 2}$
- 5.49.** $F(x, y, z) = A(x, y, z)^T$, where (a) $A = \begin{bmatrix} 1 & 2 & -3 \\ 4 & -5 & 6 \end{bmatrix}$, (b) $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$
- 5.50.** (a) $u = (2, 2)$, $k = 3$; then $F(ku) = (36, 36)$ but $kF(u) = (12, 12)$; (b) $F(0) \neq 0$;
 (c) $u = (1, 2)$, $v = (3, 4)$; then $F(u + v) = (24, 6)$ but $F(u) + F(v) = (14, 6)$;
 (d) $u = (1, 2, 3)$, $k = -2$; then $F(ku) = (2, -10)$ but $kF(u) = (-2, -10)$.
- 5.51.** $F(a, b) = (-a + 2b, -3a + b)$
- 5.52.** (a) $A = \begin{bmatrix} -17 & 5 \\ 23 & -6 \end{bmatrix}$; (b) None. $(2, -4)$ and $(-1, 2)$ are linearly dependent but not $(1, 1)$ and $(1, 3)$.
- 5.53.** $B = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}$ [Hint: Send $(0, 1)^T$ into $(0, 0)^T$.]
- 5.55.** $F(x, y) = (x^2, y^2)$
- 5.56.** (a) $13x^2 - 42xy + 34y^2 = 1$, (b) $13x^2 + 42xy + 34y^2 = 1$
- 5.57.** (a) $x^2 - 8xy + 26y^2 + 6xz - 38yz + 14z^2 = 1$, (b) $x^2 + 2xy + 3y^2 + 2xz - 8yz + 14z^2 = 1$
- 5.58.** (a) $x - y + 2z = 4$, (b) $x + 6z = 4$
- 5.61.** (a) $\dim(\text{Ker } F) = 1$, $\{(7, -2, 1)\}$; $\dim(\text{Im } F) = 2$, $\{(1, 2, 1), (0, 1, 2)\}$;
 (b) $\dim(\text{Ker } F) = 2$, $\{(-2, 1, 0, 0), (1, 0, -1, 1)\}$; $\dim(\text{Im } F) = 2$, $\{(1, 2, 1), (0, 1, 3)\}$
- 5.62.** (a) $\dim(\text{Ker } G) = 2$, $\{(1, 0, -1), (1, -1, 0)\}$; $\dim(\text{Im } G) = 1$, $\{(1, 2)\}$;
 (b) $\dim(\text{Ker } G) = 1$, $\{(1, -1, 1)\}$; $\text{Im } G = \mathbf{R}^2$, $\{(1, 0), (0, 1)\}$;
 (c) $\dim(\text{Ker } G) = 3$, $\{(-2, 1, 0, 0, 0), (1, 0, -1, 1, 0), (-5, 0, 2, 0, 1)\}$; $\dim(\text{Im } G) = 2$,
 $\{(1, 1, 3), (0, 1, 2)\}$
- 5.63.** (a) $\dim(\text{Ker } A) = 2$, $\{(4, -2, -5, 0), (1, -3, 0, 5)\}$; $\dim(\text{Im } A) = 2$, $\{(1, 2, 1), (0, 1, 1)\}$;
 (b) $\dim(\text{Ker } B) = 1$, $\{(-1, \frac{2}{3}, 1, 1)\}$; $\text{Im } B = \mathbf{R}^3$
- 5.64.** $F(x, y, z) = (x + 4y, 2x + 5y, 3x + 6y)$

- 5.65.** $F(x, y, z, t) = (x + y - z, 2x + y - t, 0)$
- 5.66.** (a) $\{1, t, t^2, \dots, t^6\}$, (b) $\{1, t, t^2, t^3\}$
- 5.68.** None, because $\dim \mathbf{R}^4 > \dim \mathbf{R}^3$.
- 5.69.** $\text{Ker } 0 = V$, $\text{Im } 0 = \{0\}$
- 5.70.** $(F + G)(x, y, z) = (y + 2z, 2x - y + z)$, $(3F - 2G)(x, y, z) = (3y - 4z, x + 2y + 3z)$
- 5.71.** (a) $(H \circ F)(x, y, z) = (x + z, 2y)$, $(H \circ G)(x, y, z) = (x - y, 4z)$; (b) not defined;
 (c) $(H \circ (F + G))(x, y, z) = (H \circ F + H \circ G)(x, y, z) = (2x - y + z, 2y + 4z)$
- 5.74.** $F(x, y) = (x, y, y)$, $G(x, y, z) = (x, y)$
- 5.75.** (a) 16, (b) 15, (c) 24
- 5.76.** (a) $v = (2, -3, 1)$; (b) $G^{-1}(at^2 + bt + c) = (b - 2c, a - b + 2c, -a + b - c)$;
 (c) H is nonsingular, but not invertible, because $\dim \mathbf{P}_2(t) > \dim \mathbf{R}^2$.
- 5.77.** $\dim U = 1$; that is, $U = K$.
- 5.78.** (a) $(F + G)(x, y) = (x, x)$; (b) $(5F - 3G)(x, y) = (5x + 8y, -3x)$; (c) $(FG)(x, y) = (x - y, 0)$;
 (d) $(GF)(x, y) = (0, x + y)$; (e) $F^2(x, y) = (x + y, 0)$ (note that $F^2 = F$); (f) $G^2(x, y) = (-x, -y)$.
 [Note that $G^2 + I = 0$; hence, G is a zero of $f(t) = t^2 + 1$.]
- 5.79.** (a) $T^{-1}(x, y) = (-3x + 2y, 2x - y)$, (b) $T^{-1}(x, y) = (-4x + 3y, -3x + 2y)$
- 5.80.** (a) $T^{-1}(x, y, z) = (x + 3y + 14z, y - 4z, z)$, (b) $T^{-1}(x, y, z) = (y + z, y, x - y - z)$
- 5.81.** (a) 49, (b) 36, (c) 144
- 5.82.** Squares: 9, 25, 36, 64, 100
- 5.83.** (a) $T(x, y) = (6x + 14y, 21x + 27y)$; (b) $T(x, y) = (0, 0)$ —that is, $f(T) = 0$

Our next theorem is analogous to Theorem 6.3 for linear operators (Problem 6.67).

THEOREM 6.12: Let S, S', S'' be bases of vector spaces V, U, W , respectively. Let $F: V \rightarrow U$ and $G: U \rightarrow W$ be linear mappings. Then

$$[G \circ F]_{S, S''} = [G]_{S', S''} [F]_{S, S'}$$

That is, relative to the appropriate bases, the matrix representation of the composition of two mappings is the matrix product of the matrix representations of the individual mappings.

Next we show how the matrix representation of a linear mapping $F: V \rightarrow U$ is affected when new bases are selected (Problem 6.67).

THEOREM 6.13: Let P be the change-of-basis matrix from a basis e to a basis e' in V , and let Q be the change-of-basis matrix from a basis f to a basis f' in U . Then, for any linear map $F: V \rightarrow U$,

$$[F]_{e', f'} = Q^{-1} [F]_{e, f} P$$

In other words, if A is the matrix representation of a linear mapping F relative to the bases e and f , and B is the matrix representation of F relative to the bases e' and f' , then

$$B = Q^{-1} A P$$

Our last theorem, proved in Problem 6.36, shows that any linear mapping from one vector space V into another vector space U can be represented by a very simple matrix. We note that this theorem is analogous to Theorem 3.18 for $m \times n$ matrices.

THEOREM 6.14: Let $F: V \rightarrow U$ be linear and, say, $\text{rank}(F) = r$. Then there exist bases of V and U such that the matrix representation of F has the form

$$A = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

where I_r is the r -square identity matrix.

The above matrix A is called the *normal* or *canonical* form of the linear map F .

SOLVED PROBLEMS

Matrix Representation of Linear Operators

6.1. Consider the linear mapping $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ defined by $F(x, y) = (3x + 4y, 2x - 5y)$ and the following bases of $\mathbf{R}^2$:

$$E = \{e_1, e_2\} = \{(1, 0), (0, 1)\} \quad \text{and} \quad S = \{u_1, u_2\} = \{(1, 2), (2, 3)\}$$

- Find the matrix A representing F relative to the basis E .
- Find the matrix B representing F relative to the basis S .
- Because E is the usual basis, the rows of A are simply the coefficients in the components of $F(x, y)$; that is, using $(a, b) = ae_1 + be_2$, we have

$$\begin{aligned} F(e_1) &= F(1, 0) = (3, 2) = 3e_1 + 2e_2 \\ F(e_2) &= F(0, 1) = (4, -5) = 4e_1 - 5e_2 \end{aligned} \quad \text{and so} \quad A = \begin{bmatrix} 3 & 4 \\ 2 & -5 \end{bmatrix}$$

Note that the coefficients of the basis vectors are written as columns in the matrix representation.

(b) First find $F(u_1)$ and write it as a linear combination of the basis vectors u_1 and u_2 . We have

$$F(u_1) = F(1, 2) = (11, -8) = x(1, 2) + y(2, 3), \quad \text{and so} \quad \begin{aligned} x + 2y &= 11 \\ 2x + 3y &= -8 \end{aligned}$$

Solve the system to obtain $x = -49$, $y = 30$. Therefore,

$$F(u_1) = -49u_1 + 30u_2$$

Next find $F(u_2)$ and write it as a linear combination of the basis vectors u_1 and u_2 . We have

$$F(u_2) = F(2, 3) = (18, -11) = x(1, 2) + y(2, 3), \quad \text{and so} \quad \begin{aligned} x + 2y &= 18 \\ 2x + 3y &= -11 \end{aligned}$$

Solve for x and y to obtain $x = -76$, $y = 47$. Hence,

$$F(u_2) = -76u_1 + 47u_2$$

Write the coefficients of u_1 and u_2 as columns to obtain $B = \begin{bmatrix} -49 & -76 \\ 30 & 47 \end{bmatrix}$

(b') Alternatively, one can first find the coordinates of an arbitrary vector (a, b) in $\mathbf{R}^2$ relative to the basis S . We have

$$(a, b) = x(1, 2) + y(2, 3) = (x + 2y, 2x + 3y), \quad \text{and so} \quad \begin{aligned} x + 2y &= a \\ 2x + 3y &= b \end{aligned}$$

Solve for x and y in terms of a and b to get $x = -3a + 2b$, $y = 2a - b$. Thus,

$$(a, b) = (-3a + 2b)u_1 + (2a - b)u_2$$

Then use the formula for (a, b) to find the coordinates of $F(u_1)$ and $F(u_2)$ relative to S :

$$\begin{aligned} F(u_1) = F(1, 2) &= (11, -8) = -49u_1 + 30u_2 \\ F(u_2) = F(2, 3) &= (18, -11) = -76u_1 + 47u_2 \end{aligned} \quad \text{and so} \quad B = \begin{bmatrix} -49 & -76 \\ 30 & 47 \end{bmatrix}$$

6.2. Consider the following linear operator G on $\mathbf{R}^2$ and basis S :

$$G(x, y) = (2x - 7y, 4x + 3y) \quad \text{and} \quad S = \{u_1, u_2\} = \{(1, 3), (2, 5)\}$$

(a) Find the matrix representation $[G]_S$ of G relative to S .

(b) Verify $[G]_S[v]_S = [G(v)]_S$ for the vector $v = (4, -3)$ in $\mathbf{R}^2$.

First find the coordinates of an arbitrary vector $v = (a, b)$ in $\mathbf{R}^2$ relative to the basis S . We have

$$\begin{bmatrix} a \\ b \end{bmatrix} = x \begin{bmatrix} 1 \\ 3 \end{bmatrix} + y \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \quad \text{and so} \quad \begin{aligned} x + 2y &= a \\ 3x + 5y &= b \end{aligned}$$

Solve for x and y in terms of a and b to get $x = -5a + 2b$, $y = 3a - b$. Thus,

$$(a, b) = (-5a + 2b)u_1 + (3a - b)u_2, \quad \text{and so} \quad [v] = [-5a + 2b, 3a - b]^T$$

(a) Using the formula for (a, b) and $G(x, y) = (2x - 7y, 4x + 3y)$, we have

$$\begin{aligned} G(u_1) &= G(1, 3) = (-19, 13) = 121u_1 - 70u_2 \\ G(u_2) &= G(2, 5) = (-31, 23) = 201u_1 - 116u_2 \end{aligned} \quad \text{and so} \quad [G]_S = \begin{bmatrix} 121 & 201 \\ -70 & -116 \end{bmatrix}$$

(We emphasize that the coefficients of u_1 and u_2 are written as columns, not rows, in the matrix representation.)

(b) Use the formula $(a, b) = (-5a + 2b)u_1 + (3a - b)u_2$ to get

$$\begin{aligned} v &= (4, -3) = -26u_1 + 15u_2 \\ G(v) &= G(4, -3) = (20, 7) = -131u_1 + 80u_2 \end{aligned}$$

Then

$$[v]_S = [-26, 15]^T \quad \text{and} \quad [G(v)]_S = [-131, 80]^T$$

Accordingly,

$$[G]_S[v]_S = \begin{bmatrix} 121 & 201 \\ -70 & -116 \end{bmatrix} \begin{bmatrix} -26 \\ 15 \end{bmatrix} = \begin{bmatrix} -131 \\ 80 \end{bmatrix} = [G(v)]_S$$

(This is expected from Theorem 6.1.)

6.3. Consider the following 2×2 matrix A and basis S of $\mathbf{R}^2$:

$$A = \begin{bmatrix} 2 & 4 \\ 5 & 6 \end{bmatrix} \quad \text{and} \quad S = \{u_1, u_2\} = \left\{ \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 3 \\ -7 \end{bmatrix} \right\}$$

The matrix A defines a linear operator on $\mathbf{R}^2$. Find the matrix B that represents the mapping A relative to the basis S .

First find the coordinates of an arbitrary vector $(a, b)^T$ with respect to the basis S . We have

$$\begin{bmatrix} a \\ b \end{bmatrix} = x \begin{bmatrix} 1 \\ -2 \end{bmatrix} + y \begin{bmatrix} 3 \\ -7 \end{bmatrix} \quad \text{or} \quad \begin{aligned} x + 3y &= a \\ -2x - 7y &= b \end{aligned}$$

Solve for x and y in terms of a and b to obtain $x = 7a + 3b$, $y = -2a - b$. Thus,

$$(a, b)^T = (7a + 3b)u_1 + (-2a - b)u_2$$

Then use the formula for $(a, b)^T$ to find the coordinates of Au_1 and Au_2 relative to the basis S :

$$\begin{aligned} Au_1 &= \begin{bmatrix} 2 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} -6 \\ -7 \end{bmatrix} = -6u_1 + 19u_2 \\ Au_2 &= \begin{bmatrix} 2 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ -7 \end{bmatrix} = \begin{bmatrix} -22 \\ -27 \end{bmatrix} = -235u_1 + 71u_2 \end{aligned}$$

Writing the coordinates as columns yields

$$B = \begin{bmatrix} -63 & -235 \\ 19 & 71 \end{bmatrix}$$

6.4. Find the matrix representation of each of the following linear operators F on $\mathbf{R}^3$ relative to the usual basis $E = \{e_1, e_2, e_3\}$ of $\mathbf{R}^3$; that is, find $[F] = [F]_E$:

(a) F defined by $F(x, y, z) = (x + 2y - 3z, 4x - 5y - 6z, 7x + 8y + 9z)$.

(b) F defined by the 3×3 matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 5 & 5 & 5 \end{bmatrix}$.

(c) F defined by $F(e_1) = (1, 3, 5)$, $F(e_2) = (2, 4, 6)$, $F(e_3) = (7, 7, 7)$. (Theorem 5.2 states that a linear map is completely defined by its action on the vectors in a basis.)

(a) Because E is the usual basis, simply write the coefficients of the components of $F(x, y, z)$ as rows:

$$[F] = \begin{bmatrix} 1 & 2 & -3 \\ 4 & -5 & -6 \\ 7 & 8 & 9 \end{bmatrix}$$

(b) Because E is the usual basis, $[F] = A$, the matrix A itself.

(c) Here

$$\begin{aligned} F(e_1) &= (1, 3, 5) = e_1 + 3e_2 + 5e_3 \\ F(e_2) &= (2, 4, 6) = 2e_1 + 4e_2 + 6e_3 \\ F(e_3) &= (7, 7, 7) = 7e_1 + 7e_2 + 7e_3 \end{aligned} \quad \text{and so} \quad [F] = \begin{bmatrix} 1 & 2 & 7 \\ 3 & 4 & 7 \\ 5 & 6 & 7 \end{bmatrix}$$

That is, the columns of $[F]$ are the images of the usual basis vectors.

6.5. Let G be the linear operator on $\mathbf{R}^3$ defined by $G(x, y, z) = (2y + z, x - 4y, 3x)$.

(a) Find the matrix representation of G relative to the basis

$$S = \{w_1, w_2, w_3\} = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$$

(b) Verify that $[G][v] = [G(v)]$ for any vector v in $\mathbf{R}^3$.

First find the coordinates of an arbitrary vector $(a, b, c) \in \mathbf{R}^3$ with respect to the basis S . Write (a, b, c) as a linear combination of w_1, w_2, w_3 using unknown scalars x, y , and z :

$$(a, b, c) = x(1, 1, 1) + y(1, 1, 0) + z(1, 0, 0) = (x + y + z, x + y, x)$$

Set corresponding components equal to each other to obtain the system of equations

$$x + y + z = a, \quad x + y = b, \quad x = c$$

Solve the system for x, y, z in terms of a, b, c to find $x = c, y = b - c, z = a - b$. Thus,

$$(a, b, c) = cw_1 + (b - c)w_2 + (a - b)w_3, \quad \text{or equivalently,} \quad [(a, b, c)] = [c, b - c, a - b]^T$$

(a) Because $G(x, y, z) = (2y + z, x - 4y, 3x)$,

$$G(w_1) = G(1, 1, 1) = (3, -3, 3) = 3w_1 - 6w_2 + 6w_3$$

$$G(w_2) = G(1, 1, 0) = (2, -3, 3) = 3w_1 - 6w_2 + 5w_3$$

$$G(w_3) = G(1, 0, 0) = (0, 1, 3) = 3w_1 - 2w_2 - w_3$$

Write the coordinates $G(w_1), G(w_2), G(w_3)$ as columns to get

$$[G] = \begin{bmatrix} 3 & 3 & 3 \\ -6 & -6 & -2 \\ 6 & 5 & -1 \end{bmatrix}$$

(b) Write $G(v)$ as a linear combination of w_1, w_2, w_3 , where $v = (a, b, c)$ is an arbitrary vector in $\mathbf{R}^3$,

$$G(v) = G(a, b, c) = (2b + c, a - 4b, 3a) = 3aw_1 + (-2a - 4b)w_2 + (-a + 6b + c)w_3$$

or equivalently,

$$[G(v)] = [3a, -2a - 4b, -a + 6b + c]^T$$

Accordingly,

$$[G][v] = \begin{bmatrix} 3 & 3 & 3 \\ -6 & -6 & -2 \\ 6 & 5 & -1 \end{bmatrix} \begin{bmatrix} c \\ b - c \\ a - b \end{bmatrix} = \begin{bmatrix} 3a \\ -2a - 4b \\ -a + 6b + c \end{bmatrix} = [G(v)]$$

6.6. Consider the following 3×3 matrix A and basis S of $\mathbf{R}^3$:

$$A = \begin{bmatrix} 1 & -2 & 1 \\ 3 & -1 & 0 \\ 1 & 4 & -2 \end{bmatrix} \quad \text{and} \quad S = \{u_1, u_2, u_3\} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right\}$$

The matrix A defines a linear operator on $\mathbf{R}^3$. Find the matrix B that represents the mapping A relative to the basis S . (Recall that A represents itself relative to the usual basis of $\mathbf{R}^3$.)

First find the coordinates of an arbitrary vector (a, b, c) in $\mathbf{R}^3$ with respect to the basis S . We have

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = x \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + z \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad \text{or} \quad \begin{aligned} x + z &= a \\ x + y + 2z &= b \\ x + y + 3z &= c \end{aligned}$$

Solve for x, y, z in terms of a, b, c to get

$$x = a + b - c, \quad y = -a + 2b - c, \quad z = c - b$$

thus, $(a, b, c)^T = (a + b - c)u_1 + (-a + 2b - c)u_2 + (c - b)u_3$

Then use the formula for $(a, b, c)^T$ to find the coordinates of Au_1, Au_2, Au_3 relative to the basis S :

$$\begin{aligned} A(u_1) &= A(1, 1, 1)^T = (0, 2, 3)^T = -u_1 + u_2 + u_3 \\ A(u_2) &= A(1, 1, 0)^T = (-1, -1, 2)^T = -4u_1 - 3u_2 + 3u_3 \\ A(u_3) &= A(1, 2, 3)^T = (0, 1, 3)^T = -2u_1 - u_2 + 2u_3 \end{aligned} \quad \text{so} \quad B = \begin{bmatrix} -1 & -4 & -2 \\ 1 & -3 & -1 \\ 1 & 3 & 2 \end{bmatrix}$$

6.7. For each of the following linear transformations (operators) L on $\mathbf{R}^2$, find the matrix A that represents L (relative to the usual basis of $\mathbf{R}^2$):

- (a) L is defined by $L(1, 0) = (2, 4)$ and $L(0, 1) = (5, 8)$.
- (b) L is the rotation in $\mathbf{R}^2$ counterclockwise by 90° .
- (c) L is the reflection in $\mathbf{R}^2$ about the line $y = -x$.

(a) Because $\{(1, 0), (0, 1)\}$ is the usual basis of $\mathbf{R}^2$, write their images under L as columns to get

$$A = \begin{bmatrix} 2 & 5 \\ 4 & 8 \end{bmatrix}$$

(b) Under the rotation L , we have $L(1, 0) = (0, 1)$ and $L(0, 1) = (-1, 0)$. Thus,

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

(c) Under the reflection L , we have $L(1, 0) = (0, -1)$ and $L(0, 1) = (-1, 0)$. Thus,

$$A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

6.8. The set $S = \{e^{3t}, te^{3t}, t^2e^{3t}\}$ is a basis of a vector space V of functions $f: \mathbf{R} \rightarrow \mathbf{R}$. Let $\mathbf{D}$ be the differential operator on V ; that is, $\mathbf{D}(f) = df/dt$. Find the matrix representation of $\mathbf{D}$ relative to the basis S .

Find the image of each basis function:

$$\begin{aligned} \mathbf{D}(e^{3t}) &= 3e^{3t} = 3(e^{3t}) + 0(te^{3t}) + 0(t^2e^{3t}) \\ \mathbf{D}(te^{3t}) &= e^{3t} + 3te^{3t} = 1(e^{3t}) + 3(te^{3t}) + 0(t^2e^{3t}) \\ \mathbf{D}(t^2e^{3t}) &= 2te^{3t} + 3t^2e^{3t} = 0(e^{3t}) + 2(te^{3t}) + 3(t^2e^{3t}) \end{aligned} \quad \text{and thus,} \quad [\mathbf{D}] = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

6.9. Prove Theorem 6.1: Let $T: V \rightarrow V$ be a linear operator, and let S be a (finite) basis of V . Then, for any vector v in V , $[T]_S[v]_S = [T(v)]_S$.

Suppose $S = \{u_1, u_2, \dots, u_n\}$, and suppose, for $i = 1, \dots, n$,

$$T(u_i) = a_{i1}u_1 + a_{i2}u_2 + \dots + a_{in}u_n = \sum_{j=1}^n a_{ij}u_j$$

Then $[T]_S$ is the n -square matrix whose j th row is

$$(a_{1j}, a_{2j}, \dots, a_{nj}) \tag{1}$$

Now suppose

$$v = k_1u_1 + k_2u_2 + \dots + k_nu_n = \sum_{i=1}^n k_iu_i$$

Writing a column vector as the transpose of a row vector, we have

$$[v]_S = [k_1, k_2, \dots, k_n]^T \tag{2}$$

Furthermore, using the linearity of T ,

$$\begin{aligned} T(v) &= T\left(\sum_{i=1}^n k_i u_i\right) = \sum_{i=1}^n k_i T(u_i) = \sum_{i=1}^n k_i \left(\sum_{j=1}^n a_{ij} u_j\right) \\ &= \sum_{j=1}^n \left(\sum_{i=1}^n a_{ij} k_i\right) u_j = \sum_{j=1}^n (a_{1j} k_1 + a_{2j} k_2 + \cdots + a_{nj} k_n) u_j \end{aligned}$$

Thus, $[T(v)]_S$ is the column vector whose j th entry is

$$a_{1j} k_1 + a_{2j} k_2 + \cdots + a_{nj} k_n \quad (3)$$

On the other hand, the j th entry of $[T]_S[v]_S$ is obtained by multiplying the j th row of $[T]_S$ by $[v]_S$ —that is (1) by (2). But the product of (1) and (2) is (3). Hence, $[T]_S[v]_S$ and $[T(v)]_S$ have the same entries. Thus, $[T]_S[v]_S = [T(v)]_S$.

6.10. Prove Theorem 6.2: Let $S = \{u_1, u_2, \dots, u_n\}$ be a basis for V over K , and let $\mathbf{M}$ be the algebra of n -square matrices over K . Then the mapping $m: A(V) \rightarrow \mathbf{M}$ defined by $m(T) = [T]_S$ is a vector space isomorphism. That is, for any $F, G \in A(V)$ and any $k \in K$, we have

(i) $[F + G] = [F] + [G]$, (ii) $[kF] = k[F]$, (iii) m is one-to-one and onto.

(i) Suppose, for $i = 1, \dots, n$,

$$F(u_i) = \sum_{j=1}^n a_{ij} u_j \quad \text{and} \quad G(u_i) = \sum_{j=1}^n b_{ij} u_j$$

Consider the matrices $A = [a_{ij}]$ and $B = [b_{ij}]$. Then $[F] = A^T$ and $[G] = B^T$. We have, for $i = 1, \dots, n$,

$$(F + G)(u_i) = F(u_i) + G(u_i) = \sum_{j=1}^n (a_{ij} + b_{ij}) u_j$$

Because $A + B$ is the matrix $(a_{ij} + b_{ij})$, we have

$$[F + G] = (A + B)^T = A^T + B^T = [F] + [G]$$

(ii) Also, for $i = 1, \dots, n$,

$$(kF)(u_i) = kF(u_i) = k \sum_{j=1}^n a_{ij} u_j = \sum_{j=1}^n (ka_{ij}) u_j$$

Because kA is the matrix (ka_{ij}) , we have

$$[kF] = (kA)^T = kA^T = k[F]$$

(iii) Finally, m is one-to-one, because a linear mapping is completely determined by its values on a basis. Also, m is onto, because matrix $A = [a_{ij}]$ in $\mathbf{M}$ is the image of the linear operator,

$$F(u_i) = \sum_{j=1}^n a_{ij} u_j, \quad i = 1, \dots, n$$

Thus, the theorem is proved.

6.11. Prove Theorem 6.3: For any linear operators $G, F \in A(V)$, $[G \circ F] = [G][F]$.

Using the notation in Problem 6.10, we have

$$\begin{aligned} (G \circ F)(u_i) &= G(F(u_i)) = G\left(\sum_{j=1}^n a_{ij} u_j\right) = \sum_{j=1}^n a_{ij} G(u_j) \\ &= \sum_{j=1}^n a_{ij} \left(\sum_{k=1}^n b_{jk} u_k\right) = \sum_{k=1}^n \left(\sum_{j=1}^n a_{ij} b_{jk}\right) u_k \end{aligned}$$

Recall that AB is the matrix $AB = [c_{ik}]$, where $c_{ik} = \sum_{j=1}^n a_{ij} b_{jk}$. Accordingly,

$$[G \circ F] = (AB)^T = B^T A^T = [G][F]$$

The theorem is proved.

6.12. Let A be the matrix representation of a linear operator T . Prove that, for any polynomial $f(t)$, we have that $f(A)$ is the matrix representation of $f(T)$. [Thus, $f(T) = 0$ if and only if $f(A) = 0$.]

Let ϕ be the mapping that sends an operator T into its matrix representation A . We need to prove that $\phi(f(T)) = f(A)$. Suppose $f(t) = a_n t^n + \cdots + a_1 t + a_0$. The proof is by induction on n , the degree of $f(t)$.

Suppose $n = 0$. Recall that $\phi(I') = I$, where I' is the identity mapping and I is the identity matrix. Thus,

$$\phi(f(T)) = \phi(a_0 I') = a_0 \phi(I') = a_0 I = f(A)$$

and so the theorem holds for $n = 0$.

Now assume the theorem holds for polynomials of degree less than n . Then, because ϕ is an algebra isomorphism,

$$\begin{aligned}\phi(f(T)) &= \phi(a_n T^n + a_{n-1} T^{n-1} + \cdots + a_1 T + a_0 I') \\ &= a_n \phi(T) \phi(T^{n-1}) + \phi(a_{n-1} T^{n-1} + \cdots + a_1 T + a_0 I') \\ &= a_n A A^{n-1} + (a_{n-1} A^{n-1} + \cdots + a_1 A + a_0 I) = f(A)\end{aligned}$$

and the theorem is proved.

Change of Basis

The coordinate vector $[v]_S$ in this section will always denote a column vector; that is,

$$[v]_S = [a_1, a_2, \dots, a_n]^T$$

6.13. Consider the following bases of $\mathbf{R}^2$:

$$E = \{e_1, e_2\} = \{(1, 0), (0, 1)\} \quad \text{and} \quad S = \{u_1, u_2\} = \{(1, 3), (1, 4)\}$$

(a) Find the change-of-basis matrix P from the usual basis E to S .

(b) Find the change-of-basis matrix Q from S back to E .

(c) Find the coordinate vector $[v]$ of $v = (5, -3)$ relative to S .

(a) Because E is the usual basis, simply write the basis vectors in S as columns: $P = \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$

(b) **Method 1.** Use the definition of the change-of-basis matrix. That is, express each vector in E as a linear combination of the vectors in S . We do this by first finding the coordinates of an arbitrary vector $v = (a, b)$ relative to S . We have

$$(a, b) = x(1, 3) + y(1, 4) = (x + y, 3x + 4y) \quad \text{or} \quad \begin{cases} x + y = a \\ 3x + 4y = b \end{cases}$$

Solve for x and y to obtain $x = 4a - b$, $y = -3a + b$. Thus,

$$v = (4a - b)u_1 + (-3a + b)u_2 \quad \text{and} \quad [v]_S = [(a, b)]_S = [4a - b, -3a + b]^T$$

Using the above formula for $[v]_S$ and writing the coordinates of the e_i as columns yields

$$\begin{aligned}e_1 = (1, 0) &= 4u_1 - 3u_2 \\ e_2 = (0, 1) &= -u_1 + u_2\end{aligned} \quad \text{and} \quad Q = \begin{bmatrix} 4 & -1 \\ -3 & 1 \end{bmatrix}$$

Method 2. Because $Q = P^{-1}$, find P^{-1} , say by using the formula for the inverse of a 2×2 matrix. Thus,

$$P^{-1} = \begin{bmatrix} 4 & -1 \\ -3 & 1 \end{bmatrix}$$

(c) **Method 1.** Write v as a linear combination of the vectors in S , say by using the above formula for $v = (a, b)$. We have $v = (5, -3) = 23u_1 - 18u_2$, and so $[v]_S = [23, -18]^T$.

Method 2. Use, from Theorem 6.6, the fact that $[v]_S = P^{-1}[v]_E$ and the fact that $[v]_E = [5, -3]^T$:

$$[v]_S = P^{-1}[v]_E = \begin{bmatrix} 4 & -1 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ -3 \end{bmatrix} = \begin{bmatrix} 23 \\ -18 \end{bmatrix}$$

6.14. The vectors $u_1 = (1, 2, 0)$, $u_2 = (1, 3, 2)$, $u_3 = (0, 1, 3)$ form a basis S of $\mathbf{R}^3$. Find

(a) The change-of-basis matrix P from the usual basis $E = \{e_1, e_2, e_3\}$ to S .

(b) The change-of-basis matrix Q from S back to E .

(a) Because E is the usual basis, simply write the basis vectors of S as columns: $P = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 1 \\ 0 & 2 & 3 \end{bmatrix}$

(b) **Method 1.** Express each basis vector of E as a linear combination of the basis vectors of S by first finding the coordinates of an arbitrary vector $v = (a, b, c)$ relative to the basis S . We have

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = x \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + y \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} + z \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix} \quad \text{or} \quad \begin{aligned} x + y &= a \\ 2x + 3y + z &= b \\ 2y + 3z &= c \end{aligned}$$

Solve for x, y, z to get $x = 7a - 3b + c$, $y = -6a + 3b - c$, $z = 4a - 2b + c$. Thus,

$$v = (a, b, c) = (7a - 3b + c)u_1 + (-6a + 3b - c)u_2 + (4a - 2b + c)u_3$$

or $[v]_S = [(a, b, c)]_S = [7a - 3b + c, -6a + 3b - c, 4a - 2b + c]^T$

Using the above formula for $[v]_S$ and then writing the coordinates of the e_i as columns yields

$$\begin{aligned} e_1 = (1, 0, 0) &= 7u_1 - 6u_2 + 4u_3 \\ e_2 = (0, 1, 0) &= -3u_1 + 3u_2 - 2u_3 \\ e_3 = (0, 0, 1) &= u_1 - u_2 + u_3 \end{aligned} \quad \text{and} \quad Q = \begin{bmatrix} 7 & -3 & 1 \\ -6 & 3 & -1 \\ 4 & -2 & 1 \end{bmatrix}$$

Method 2. Find P^{-1} by row reducing $M = [P, I]$ to the form $[I, P^{-1}]$:

$$\begin{aligned} M &= \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 2 & 3 & 1 & 0 & 1 & 0 \\ 0 & 2 & 3 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 2 & 3 & 0 & 0 & 1 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 0 & 1 & 4 & -2 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 7 & -3 & 1 \\ 0 & 1 & 0 & -6 & 3 & -1 \\ 0 & 0 & 1 & 4 & -2 & 1 \end{array} \right] = [I, P^{-1}] \end{aligned}$$

$$\text{Thus, } Q = P^{-1} = \begin{bmatrix} 7 & -3 & 1 \\ -6 & 3 & -1 \\ 4 & -2 & 1 \end{bmatrix}.$$

6.15. Suppose the x -axis and y -axis in the plane $\mathbf{R}^2$ are rotated counterclockwise 45° so that the new x' -axis and y' -axis are along the line $y = x$ and the line $y = -x$, respectively.

(a) Find the change-of-basis matrix P .

(b) Find the coordinates of the point $A(5, 6)$ under the given rotation.

(a) The unit vectors in the direction of the new x' - and y' -axes are

$$u_1 = (\tfrac{1}{2}\sqrt{2}, \tfrac{1}{2}\sqrt{2}) \quad \text{and} \quad u_2 = (-\tfrac{1}{2}\sqrt{2}, \tfrac{1}{2}\sqrt{2})$$

(The unit vectors in the direction of the original x and y axes are the usual basis of $\mathbf{R}^2$.) Thus, write the coordinates of u_1 and u_2 as columns to obtain

$$P = \begin{bmatrix} \frac{1}{2}\sqrt{2} & -\frac{1}{2}\sqrt{2} \\ \frac{1}{2}\sqrt{2} & \frac{1}{2}\sqrt{2} \end{bmatrix}$$

(b) Multiply the coordinates of the point by P^{-1} :

$$\begin{bmatrix} \frac{1}{2}\sqrt{2} & \frac{1}{2}\sqrt{2} \\ -\frac{1}{2}\sqrt{2} & \frac{1}{2}\sqrt{2} \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} \frac{11}{2}\sqrt{2} \\ \frac{1}{2}\sqrt{2} \end{bmatrix}$$

(Because P is orthogonal, P^{-1} is simply the transpose of P .)

6.16. The vectors $u_1 = (1, 1, 0)$, $u_2 = (0, 1, 1)$, $u_3 = (1, 2, 2)$ form a basis S of $\mathbf{R}^3$. Find the coordinates of an arbitrary vector $v = (a, b, c)$ relative to the basis S .

Method 1. Express v as a linear combination of u_1, u_2, u_3 using unknowns x, y, z . We have

$$(a, b, c) = x(1, 1, 0) + y(0, 1, 1) + z(1, 2, 2) = (x + z, x + y + 2z, y + 2z)$$

this yields the system

$$\begin{array}{rcl} x + z = a & \text{or} & x + z = a \\ x + y + 2z = b & \text{or} & y + z = -a + b \\ y + 2z = c & \text{or} & y + 2z = c \end{array} \quad \text{or} \quad \begin{array}{rcl} x + z = a & & x + z = a \\ y + z = -a + b & & y + z = -a + b \\ z = a - b + c & & z = a - b + c \end{array}$$

Solving by back-substitution yields $x = b - c$, $y = -2a + 2b - c$, $z = a - b + c$. Thus,

$$[v]_S = [b - c, -2a + 2b - c, a - b + c]^T$$

Method 2. Find P^{-1} by row reducing $M = [P, I]$ to the form $[I, P^{-1}]$, where P is the change-of-basis matrix from the usual basis E to S or, in other words, the matrix whose columns are the basis vectors of S .

We have

$$\begin{aligned} M &= \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 2 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & -2 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array} \right] = [I, P^{-1}] \\ \text{Thus, } P^{-1} &= \begin{bmatrix} 0 & 1 & -1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{bmatrix} \text{ and } [v]_S = P^{-1}[v]_E = \begin{bmatrix} 0 & 1 & -1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} b - c \\ -2a + 2b - c \\ a - b + c \end{bmatrix} \end{aligned}$$

6.17. Consider the following bases of $\mathbf{R}^2$:

$$S = \{u_1, u_2\} = \{(1, -2), (3, -4)\} \quad \text{and} \quad S' = \{v_1, v_2\} = \{(1, 3), (3, 8)\}$$

- Find the coordinates of $v = (a, b)$ relative to the basis S .
- Find the change-of-basis matrix P from S to S' .
- Find the coordinates of $v = (a, b)$ relative to the basis S' .
- Find the change-of-basis matrix Q from S' back to S .
- Verify $Q = P^{-1}$.
- Show that, for any vector $v = (a, b)$ in $\mathbf{R}^2$, $P^{-1}[v]_S = [v]_{S'}$. (See Theorem 6.6.)

(a) Let $v = xu_1 + yu_2$ for unknowns x and y ; that is,

$$\begin{bmatrix} a \\ b \end{bmatrix} = x \begin{bmatrix} 1 \\ -2 \end{bmatrix} + y \begin{bmatrix} 3 \\ -4 \end{bmatrix} \quad \text{or} \quad \begin{array}{rcl} x + 3y & = & a \\ -2x - 4y & = & b \end{array} \quad \text{or} \quad \begin{array}{rcl} x + 3y & = & a \\ 2y & = & 2a + b \end{array}$$

Solve for x and y in terms of a and b to get $x = -2a - \frac{3}{2}b$ and $y = a + \frac{1}{2}b$. Thus,

$$(a, b) = (-2a - \frac{3}{2}b)u_1 + (a + \frac{1}{2}b)u_2 \quad \text{or} \quad [(a, b)]_S = [-2a - \frac{3}{2}b, a + \frac{1}{2}b]^T$$

(b) Use part (a) to write each of the basis vectors v_1 and v_2 of S' as a linear combination of the basis vectors u_1 and u_2 of S ; that is,

$$v_1 = (1, 3) = (-2 - \frac{9}{2})u_1 + (1 + \frac{3}{2})u_2 = -\frac{13}{2}u_1 + \frac{5}{2}u_2$$

$$v_2 = (3, 8) = (-6 - 12)u_1 + (3 + 4)u_2 = -18u_1 + 7u_2$$

Then P is the matrix whose columns are the coordinates of v_1 and v_2 relative to the basis S ; that is,

$$P = \begin{bmatrix} -\frac{13}{2} & -18 \\ \frac{5}{2} & 7 \end{bmatrix}$$

(c) Let $v = xv_1 + yv_2$ for unknown scalars x and y :

$$\begin{bmatrix} a \\ b \end{bmatrix} = x \begin{bmatrix} 1 \\ 3 \end{bmatrix} + y \begin{bmatrix} 3 \\ 8 \end{bmatrix} \quad \text{or} \quad \begin{array}{l} x + 3y = a \\ 3x + 8y = b \end{array} \quad \text{or} \quad \begin{array}{l} x + 3y = a \\ -y = b - 3a \end{array}$$

Solve for x and y to get $x = -8a + 3b$ and $y = 3a - b$. Thus,

$$(a, b) = (-8a + 3b)v_1 + (3a - b)v_2 \quad \text{or} \quad [(a, b)]_{S'} = [-8a + 3b, \quad 3a - b]^T$$

(d) Use part (c) to express each of the basis vectors u_1 and u_2 of S as a linear combination of the basis vectors v_1 and v_2 of S' :

$$u_1 = (1, -2) = (-8 - 6)v_1 + (3 + 2)v_2 = -14v_1 + 5v_2$$

$$u_2 = (3, -4) = (-24 - 12)v_1 + (9 + 4)v_2 = -36v_1 + 13v_2$$

Write the coordinates of u_1 and u_2 relative to S' as columns to obtain $Q = \begin{bmatrix} -14 & -36 \\ 5 & 13 \end{bmatrix}$.

$$(e) \quad QP = \begin{bmatrix} -14 & -36 \\ 5 & 13 \end{bmatrix} \begin{bmatrix} -\frac{13}{2} & -18 \\ \frac{5}{2} & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

(f) Use parts (a), (c), and (d) to obtain

$$P^{-1}[v]_S = Q[v]_{S'} = \begin{bmatrix} -14 & -36 \\ 5 & 13 \end{bmatrix} \begin{bmatrix} -2a - \frac{3}{2}b \\ a + \frac{1}{2}b \end{bmatrix} = \begin{bmatrix} -8a + 3b \\ 3a - b \end{bmatrix} = [v]_{S'}$$

6.18. Suppose P is the change-of-basis matrix from a basis $\{u_i\}$ to a basis $\{w_i\}$, and suppose Q is the change-of-basis matrix from the basis $\{w_i\}$ back to $\{u_i\}$. Prove that P is invertible and that $Q = P^{-1}$.

Suppose, for $i = 1, 2, \dots, n$, that

$$w_i = a_{i1}u_1 + a_{i2}u_2 + \dots + a_{in}u_n = \sum_{j=1}^n a_{ij}u_j \quad (1)$$

and, for $j = 1, 2, \dots, n$,

$$u_j = b_{j1}w_1 + b_{j2}w_2 + \dots + b_{jn}w_n = \sum_{k=1}^n b_{jk}w_k \quad (2)$$

Let $A = [a_{ij}]$ and $B = [b_{jk}]$. Then $P = A^T$ and $Q = B^T$. Substituting (2) into (1) yields

$$w_i = \sum_{j=1}^n a_{ij} \left(\sum_{k=1}^n b_{jk}w_k \right) = \sum_{k=1}^n \left(\sum_{j=1}^n a_{ij}b_{jk} \right) w_k$$

Because $\{w_i\}$ is a basis, $\sum a_{ij}b_{jk} = \delta_{ik}$, where δ_{ik} is the Kronecker delta; that is, $\delta_{ik} = 1$ if $i = k$ but $\delta_{ik} = 0$ if $i \neq k$. Suppose $AB = [c_{ik}]$. Then $c_{ik} = \delta_{ik}$. Accordingly, $AB = I$, and so

$$QP = B^T A^T = (AB)^T = I^T = I$$

Thus, $Q = P^{-1}$.

6.19. Consider a finite sequence of vectors $S = \{u_1, u_2, \dots, u_n\}$. Let S' be the sequence of vectors obtained from S by one of the following “elementary operations”:

- (1) Interchange two vectors.
- (2) Multiply a vector by a nonzero scalar.
- (3) Add a multiple of one vector to another vector.

Show that S and S' span the same subspace $\mathcal{W}$. Also, show that S' is linearly independent if and only if S is linearly independent.

Observe that, for each operation, the vectors S' are linear combinations of vectors in S . Also, because each operation has an inverse of the same type, each vector in S is a linear combination of vectors in S' . Thus, S and S' span the same subspace W . Moreover, S' is linearly independent if and only if $\dim W = n$, and this is true if and only if S is linearly independent.

- 6.20.** Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be row equivalent $m \times n$ matrices over a field K , and let $v_1, v_2, \dots, v_n$ be any vectors in a vector space V over K . For $i = 1, 2, \dots, m$, let u_i and w_i be defined by

$$u_i = a_{i1}v_1 + a_{i2}v_2 + \cdots + a_{in}v_n \quad \text{and} \quad w_i = b_{i1}v_1 + b_{i2}v_2 + \cdots + b_{in}v_n$$

Show that $\{u_i\}$ and $\{w_i\}$ span the same subspace of V .

Applying an “elementary operation” of Problem 6.19 to $\{u_i\}$ is equivalent to applying an elementary row operation to the matrix A . Because A and B are row equivalent, B can be obtained from A by a sequence of elementary row operations. Hence, $\{w_i\}$ can be obtained from $\{u_i\}$ by the corresponding sequence of operations. Accordingly, $\{u_i\}$ and $\{w_i\}$ span the same space.

- 6.21.** Suppose $u_1, u_2, \dots, u_n$ belong to a vector space V over a field K , and suppose $P = [a_{ij}]$ is an n -square matrix over K . For $i = 1, 2, \dots, n$, let $v_i = a_{i1}u_1 + a_{i2}u_2 + \cdots + a_{in}u_n$.

- Suppose P is invertible. Show that $\{u_i\}$ and $\{v_i\}$ span the same subspace of V . Hence, $\{u_i\}$ is linearly independent if and only if $\{v_i\}$ is linearly independent.
 - Suppose P is singular (not invertible). Show that $\{v_i\}$ is linearly dependent.
 - Suppose $\{v_i\}$ is linearly independent. Show that P is invertible.
- Because P is invertible, it is row equivalent to the identity matrix I . Hence, by Problem 6.19, $\{v_i\}$ and $\{u_i\}$ span the same subspace of V . Thus, one is linearly independent if and only if the other is linearly independent.
 - Because P is not invertible, it is row equivalent to a matrix with a zero row. This means $\{v_i\}$ spans a subspace that has a spanning set with less than n elements. Thus, $\{v_i\}$ is linearly dependent.
 - This is the contrapositive of the statement of part (b), and so it follows from part (b).

- 6.22.** Prove Theorem 6.6: Let P be the change-of-basis matrix from a basis S to a basis S' in a vector space V . Then, for any vector $v \in V$, we have $P[v]_{S'} = [v]_S$, and hence, $P^{-1}[v]_S = [v]_{S'}$.

Suppose $S = \{u_1, \dots, u_n\}$ and $S' = \{w_1, \dots, w_n\}$, and suppose, for $i = 1, \dots, n$,

$$w_i = a_{i1}u_1 + a_{i2}u_2 + \cdots + a_{in}u_n = \sum_{j=1}^n a_{ij}u_j$$

Then P is the n -square matrix whose j th row is

$$(a_{1j}, a_{2j}, \dots, a_{nj}) \quad (1)$$

Also suppose $v = k_1w_1 + k_2w_2 + \cdots + k_nw_n = \sum_{i=1}^n k_iw_i$. Then

$$[v]_{S'} = [k_1, k_2, \dots, k_n]^T \quad (2)$$

Substituting for w_i in the equation for v , we obtain

$$\begin{aligned} v &= \sum_{i=1}^n k_iw_i = \sum_{i=1}^n k_i \left(\sum_{j=1}^n a_{ij}u_j \right) = \sum_{j=1}^n \left(\sum_{i=1}^n a_{ij}k_i \right) u_j \\ &= \sum_{j=1}^n (a_{1j}k_1 + a_{2j}k_2 + \cdots + a_{nj}k_n) u_j \end{aligned}$$

Accordingly, $[v]_S$ is the column vector whose j th entry is

$$a_{1j}k_1 + a_{2j}k_2 + \cdots + a_{nj}k_n \quad (3)$$

On the other hand, the j th entry of $P[v]_{S'}$ is obtained by multiplying the j th row of P by $[v]_{S'}$ —that is, (1) by (2). However, the product of (1) and (2) is (3). Hence, $P[v]_{S'}$ and $[v]_S$ have the same entries. Thus, $P[v]_{S'} = [v]_S$, as claimed.

Furthermore, multiplying the above by P^{-1} gives $P^{-1}[v]_S = P^{-1}P[v]_{S'} = [v]_{S'}$.

Linear Operators and Change of Basis

6.23. Consider the linear transformation F on $\mathbf{R}^2$ defined by $F(x, y) = (5x - y, 2x + y)$ and the following bases of $\mathbf{R}^2$:

$$E = \{e_1, e_2\} = \{(1, 0), (0, 1)\} \quad \text{and} \quad S = \{u_1, u_2\} = \{(1, 4), (2, 7)\}$$

- (a) Find the change-of-basis matrix P from E to S and the change-of-basis matrix Q from S back to E .
 (b) Find the matrix A that represents F in the basis E .
 (c) Find the matrix B that represents F in the basis S .
 (a) Because E is the usual basis, simply write the vectors in S as columns to obtain the change-of-basis matrix P . Recall, also, that $Q = P^{-1}$. Thus,

$$P = \begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix} \quad \text{and} \quad Q = P^{-1} = \begin{bmatrix} -7 & 2 \\ 4 & -1 \end{bmatrix}$$

- (b) Write the coefficients of x and y in $F(x, y) = (5x - y, 2x + y)$ as rows to get

$$A = \begin{bmatrix} 5 & -1 \\ 2 & 1 \end{bmatrix}$$

- (c) **Method 1.** Find the coordinates of $F(u_1)$ and $F(u_2)$ relative to the basis S . This may be done by first finding the coordinates of an arbitrary vector (a, b) in $\mathbf{R}^2$ relative to the basis S . We have

$$(a, b) = x(1, 4) + y(2, 7) = (x + 2y, 4x + 7y), \quad \text{and so} \quad \begin{aligned} x + 2y &= a \\ 4x + 7y &= b \end{aligned}$$

Solve for x and y in terms of a and b to get $x = -7a + 2b$, $y = 4a - b$. Then

$$(a, b) = (-7a + 2b)u_1 + (4a - b)u_2$$

Now use the formula for (a, b) to obtain

$$\begin{aligned} F(u_1) &= F(1, 4) = (1, 6) = 5u_1 - 2u_2 \\ F(u_2) &= F(2, 7) = (3, 11) = u_1 + u_2 \end{aligned} \quad \text{and so} \quad B = \begin{bmatrix} 5 & 1 \\ -2 & 1 \end{bmatrix}$$

Method 2. By Theorem 6.7, $B = P^{-1}AP$. Thus,

$$B = P^{-1}AP = \begin{bmatrix} -7 & 2 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} 5 & -1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ -2 & 1 \end{bmatrix}$$

6.24. Let $A = \begin{bmatrix} 2 & 3 \\ 4 & -1 \end{bmatrix}$. Find the matrix B that represents the linear operator A relative to the basis $S = \{u_1, u_2\} = \{[1, 3]^T, [2, 5]^T\}$. [Recall A defines a linear operator $A: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ relative to the usual basis E of $\mathbf{R}^2$].

Method 1. Find the coordinates of $A(u_1)$ and $A(u_2)$ relative to the basis S by first finding the coordinates of an arbitrary vector $[a, b]^T$ in $\mathbf{R}^2$ relative to the basis S . By Problem 6.2,

$$[a, b]^T = (-5a + 2b)u_1 + (3a - b)u_2$$

Using the formula for $[a, b]^T$, we obtain

$$A(u_1) = \begin{bmatrix} 2 & 3 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 11 \\ 1 \end{bmatrix} = -53u_1 + 32u_2$$

and

$$A(u_2) = \begin{bmatrix} 2 & 3 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 19 \\ 3 \end{bmatrix} = -89u_1 + 54u_2$$

Thus,

$$B = \begin{bmatrix} -53 & -89 \\ 32 & 54 \end{bmatrix}$$

Method 2. Use $B = P^{-1}AP$, where P is the change-of-basis matrix from the usual basis E to S . Thus, simply write the vectors in S (as columns) to obtain the change-of-basis matrix P and then use the formula

for P^{-1} . This gives

$$P = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \quad \text{and} \quad P^{-1} = \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}$$

Then
$$B = P^{-1}AP = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} -53 & -89 \\ 32 & 54 \end{bmatrix}$$

6.25. Let $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & -4 \\ 1 & -2 & 2 \end{bmatrix}$. Find the matrix B that represents the linear operator A relative to the basis

$$S = \{u_1, u_2, u_3\} = \{[1, 1, 0]^T, [0, 1, 1]^T, [1, 2, 2]^T\}$$

[Recall A that defines a linear operator $A: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ relative to the usual basis E of $\mathbf{R}^3$.]

Method 1. Find the coordinates of $A(u_1)$, $A(u_2)$, $A(u_3)$ relative to the basis S by first finding the coordinates of an arbitrary vector $v = (a, b, c)$ in $\mathbf{R}^3$ relative to the basis S . By Problem 6.16,

$$[v]_S = (b - c)u_1 + (-2a + 2b - c)u_2 + (a - b + c)u_3$$

Using this formula for $[a, b, c]^T$, we obtain

$$\begin{aligned} A(u_1) &= [4, 7, -1]^T = 8u_1 + 7u_2 - 5u_3, & A(u_2) &= [4, 1, 0]^T = u_1 - 6u_2 + 3u_3 \\ A(u_3) &= [9, 4, 1]^T = 3u_1 - 11u_2 + 6u_3 \end{aligned}$$

Writing the coefficients of u_1, u_2, u_3 as columns yields

$$B = \begin{bmatrix} 8 & 1 & 3 \\ 7 & -6 & -11 \\ -5 & 3 & 6 \end{bmatrix}$$

Method 2. Use $B = P^{-1}AP$, where P is the change-of-basis matrix from the usual basis E to S . The matrix P (whose columns are simply the vectors in S) and P^{-1} appear in Problem 6.16. Thus,

$$B = P^{-1}AP = \begin{bmatrix} 0 & 1 & -1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & -4 \\ 1 & -2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 8 & 1 & 3 \\ 7 & -6 & -11 \\ -5 & 3 & 6 \end{bmatrix}$$

6.26. Prove Theorem 6.7: Let P be the change-of-basis matrix from a basis S to a basis S' in a vector space V . Then, for any linear operator T on V , $[T]_{S'} = P^{-1}[T]_S P$.

Let v be a vector in V . Then, by Theorem 6.6, $P[v]_{S'} = [v]_S$. Therefore,

$$P^{-1}[T]_S P[v]_{S'} = P^{-1}[T]_S [v]_S = P^{-1}[T(v)]_S = [T(v)]_{S'}$$

But $[T]_{S'}[v]_{S'} = [T(v)]_{S'}$. Hence,

$$P^{-1}[T]_S P[v]_{S'} = [T]_{S'}[v]_{S'}$$

Because the mapping $v \mapsto [v]_{S'}$ is onto K^n , we have $P^{-1}[T]_S PX = [T]_{S'}X$ for every $X \in K^n$. Thus, $P^{-1}[T]_S P = [T]_{S'}$, as claimed.

Similarity of Matrices

6.27. Let $A = \begin{bmatrix} 4 & -2 \\ 3 & 6 \end{bmatrix}$ and $P = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

(a) Find $B = P^{-1}AP$. (b) Verify $\text{tr}(B) = \text{tr}(A)$. (c) Verify $\det(B) = \det(A)$.

(a) First find P^{-1} using the formula for the inverse of a 2×2 matrix. We have

$$P^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

Then

$$B = P^{-1}AP = \begin{bmatrix} -2 & 1 \\ 3 & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 4 & -2 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 25 & 30 \\ -\frac{27}{2} & -15 \end{bmatrix}$$

- (b) $\text{tr}(A) = 4 + 6 = 10$ and $\text{tr}(B) = 25 - 15 = 10$. Hence, $\text{tr}(B) = \text{tr}(A)$.
 (c) $\det(A) = 24 + 6 = 30$ and $\det(B) = -375 + 405 = 30$. Hence, $\det(B) = \det(A)$.

6.28. Find the trace of each of the linear transformations F on $\mathbf{R}^3$ in Problem 6.4.

Find the trace (sum of the diagonal elements) of any matrix representation of F such as the matrix representation $[F] = [F]_E$ of F relative to the usual basis E given in Problem 6.4.

- (a) $\text{tr}(F) = \text{tr}([F]) = 1 - 5 + 9 = 5$.
 (b) $\text{tr}(F) = \text{tr}([F]) = 1 + 3 + 5 = 9$.
 (c) $\text{tr}(F) = \text{tr}([F]) = 1 + 4 + 7 = 12$.

6.29. Write $A \approx B$ if A is similar to B —that is, if there exists an invertible matrix P such that $A = P^{-1}BP$. Prove that $\approx$ is an equivalence relation (on square matrices); that is,

- (a) $A \approx A$, for every A . (b) If $A \approx B$, then $B \approx A$.
 (c) If $A \approx B$ and $B \approx C$, then $A \approx C$.
 (a) The identity matrix I is invertible, and $I^{-1} = I$. Because $A = I^{-1}AI$, we have $A \approx A$.
 (b) Because $A \approx B$, there exists an invertible matrix P such that $A = P^{-1}BP$. Hence, $B = PAP^{-1} = (P^{-1})^{-1}AP$ and P^{-1} is also invertible. Thus, $B \approx A$.
 (c) Because $A \approx B$, there exists an invertible matrix P such that $A = P^{-1}BP$, and as $B \approx C$, there exists an invertible matrix Q such that $B = Q^{-1}CQ$. Thus,

$$A = P^{-1}BP = P^{-1}(Q^{-1}CQ)P = (P^{-1}Q^{-1})C(QP) = (QP)^{-1}C(QP)$$

and QP is also invertible. Thus, $A \approx C$.

6.30. Suppose B is similar to A , say $B = P^{-1}AP$. Prove

- (a) $B^n = P^{-1}A^nP$, and so B^n is similar to A^n .
 (b) $f(B) = P^{-1}f(A)P$, for any polynomial $f(x)$, and so $f(B)$ is similar to $f(A)$.
 (c) B is a root of a polynomial $g(x)$ if and only if A is a root of $g(x)$.
 (a) The proof is by induction on n . The result holds for $n = 1$ by hypothesis. Suppose $n > 1$ and the result holds for $n - 1$. Then

$$B^n = BB^{n-1} = (P^{-1}AP)(P^{-1}A^{n-1}P) = P^{-1}A^nP$$

- (b) Suppose $f(x) = a_nx^n + \cdots + a_1x + a_0$. Using the left and right distributive laws and part (a), we have

$$\begin{aligned} P^{-1}f(A)P &= P^{-1}(a_nA^n + \cdots + a_1A + a_0I)P \\ &= P^{-1}(a_nA^n)P + \cdots + P^{-1}(a_1A)P + P^{-1}(a_0I)P \\ &= a_n(P^{-1}A^nP) + \cdots + a_1(P^{-1}AP) + a_0(P^{-1}IP) \\ &= a_nB^n + \cdots + a_1B + a_0I = f(B) \end{aligned}$$

- (c) By part (b), $g(B) = 0$ if and only if $P^{-1}g(A)P = 0$ if and only if $g(A) = P0P^{-1} = 0$.

Matrix Representations of General Linear Mappings

6.31. Let $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ be the linear map defined by $F(x, y, z) = (3x + 2y - 4z, x - 5y + 3z)$.

- (a) Find the matrix of F in the following bases of $\mathbf{R}^3$ and $\mathbf{R}^2$:

$$S = \{w_1, w_2, w_3\} = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\} \quad \text{and} \quad S' = \{u_1, u_2\} = \{(1, 3), (2, 5)\}$$

(b) Verify Theorem 6.10: The action of F is preserved by its matrix representation; that is, for any v in $\mathbf{R}^3$, we have $[F]_{S,S'}[v]_S = [F(v)]_{S'}$.

(a) From Problem 6.2, $(a, b) = (-5a + 2b)u_1 + (3a - b)u_2$. Thus,

$$\begin{aligned} F(w_1) &= F(1, 1, 1) = (1, -1) = -7u_1 + 4u_2 \\ F(w_2) &= F(1, 1, 0) = (5, -4) = -33u_1 + 19u_2 \\ F(w_3) &= F(1, 0, 0) = (3, 1) = -13u_1 + 8u_2 \end{aligned}$$

Write the coordinates of $F(w_1), F(w_2), F(w_3)$ as columns to get

$$[F]_{S,S'} = \begin{bmatrix} -7 & -33 & 13 \\ 4 & 19 & 8 \end{bmatrix}$$

(b) If $v = (x, y, z)$, then, by Problem 6.5, $v = zw_1 + (y - z)w_2 + (x - y)w_3$. Also,

$$F(v) = (3x + 2y - 4z, x - 5y + 3z) = (-13x - 20y + 26z)u_1 + (8x + 11y - 15z)u_2$$

$$\text{Hence, } [v]_S = (z, y - z, x - y)^T \quad \text{and} \quad [F(v)]_{S'} = \begin{bmatrix} -13x - 20y + 26z \\ 8x + 11y - 15z \end{bmatrix}$$

$$\text{Thus, } [F]_{S,S'}[v]_S = \begin{bmatrix} -7 & -33 & 13 \\ 4 & 19 & 8 \end{bmatrix} \begin{bmatrix} z \\ y - z \\ x - y \end{bmatrix} = \begin{bmatrix} -13x - 20y + 26z \\ 8x + 11y - 15z \end{bmatrix} = [F(v)]_{S'}$$

6.32. Let $F: \mathbf{R}^n \rightarrow \mathbf{R}^m$ be the linear mapping defined as follows:

$$F(x_1, x_2, \dots, x_n) = (a_{11}x_1 + \dots + a_{1n}x_n, a_{21}x_1 + \dots + a_{2n}x_n, \dots, a_{m1}x_1 + \dots + a_{mn}x_n)$$

(a) Show that the rows of the matrix $[F]$ representing F relative to the usual bases of $\mathbf{R}^n$ and $\mathbf{R}^m$ are the coefficients of the x_i in the components of $F(x_1, \dots, x_n)$.

(b) Find the matrix representation of each of the following linear mappings relative to the usual basis of $\mathbf{R}^n$:

- (i) $F: \mathbf{R}^2 \rightarrow \mathbf{R}^3$ defined by $F(x, y) = (3x - y, 2x + 4y, 5x - 6y)$.
- (ii) $F: \mathbf{R}^4 \rightarrow \mathbf{R}^2$ defined by $F(x, y, s, t) = (3x - 4y + 2s - 5t, 5x + 7y - s - 2t)$.
- (iii) $F: \mathbf{R}^3 \rightarrow \mathbf{R}^4$ defined by $F(x, y, z) = (2x + 3y - 8z, x + y + z, 4x - 5z, 6y)$.

(a) We have

$$\begin{aligned} F(1, 0, \dots, 0) &= (a_{11}, a_{21}, \dots, a_{m1}) \\ F(0, 1, \dots, 0) &= (a_{12}, a_{22}, \dots, a_{m2}) \\ &\vdots \\ F(0, 0, \dots, 1) &= (a_{1n}, a_{2n}, \dots, a_{mn}) \end{aligned} \quad \text{and thus, } [F] = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

(b) By part (a), we need only look at the coefficients of the unknown $x, y, \dots$ in $F(x, y, \dots)$. Thus,

$$\begin{aligned} \text{(i)} \quad [F] &= \begin{bmatrix} 3 & -1 \\ 2 & 4 \\ 5 & -6 \end{bmatrix}, & \text{(ii)} \quad [F] &= \begin{bmatrix} 3 & -4 & 2 & -5 \\ 5 & 7 & -1 & -2 \end{bmatrix}, & \text{(iii)} \quad [F] &= \begin{bmatrix} 2 & 3 & -8 \\ 1 & 1 & 1 \\ 4 & 0 & -5 \\ 0 & 6 & 0 \end{bmatrix} \end{aligned}$$

6.33. Let $A = \begin{bmatrix} 2 & 5 & -3 \\ 1 & -4 & 7 \end{bmatrix}$. Recall that A determines a mapping $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by $F(v) = Av$, where vectors are written as columns. Find the matrix $[F]$ that represents the mapping relative to the following bases of $\mathbf{R}^3$ and $\mathbf{R}^2$:

- (a) The usual bases of $\mathbf{R}^3$ and of $\mathbf{R}^2$.
- (b) $S = \{w_1, w_2, w_3\} = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$ and $S' = \{u_1, u_2\} = \{(1, 3), (2, 5)\}$.
- (a) Relative to the usual bases, $[F]$ is the matrix A .

(b) From Problem 9.2, $(a, b) = (-5a + 2b)u_1 + (3a - b)u_2$. Thus,

$$\begin{aligned} F(w_1) &= \begin{bmatrix} 2 & 5 & -3 \\ 1 & -4 & 7 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix} = -12u_1 + 8u_2 \\ F(w_2) &= \begin{bmatrix} 2 & 5 & -3 \\ 1 & -4 & 7 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 7 \\ -3 \end{bmatrix} = -41u_1 + 24u_2 \\ F(w_3) &= \begin{bmatrix} 2 & 5 & -3 \\ 1 & -4 & 7 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} = -8u_1 + 5u_2 \end{aligned}$$

Writing the coefficients of $F(w_1)$, $F(w_2)$, $F(w_3)$ as columns yields $[F] = \begin{bmatrix} -12 & -41 & -8 \\ 8 & 24 & 5 \end{bmatrix}$.

6.34. Consider the linear transformation T on $\mathbf{R}^2$ defined by $T(x, y) = (2x - 3y, x + 4y)$ and the following bases of $\mathbf{R}^2$:

$$E = \{e_1, e_2\} = \{(1, 0), (0, 1)\} \quad \text{and} \quad S = \{u_1, u_2\} = \{(1, 3), (2, 5)\}$$

- (a) Find the matrix A representing T relative to the bases E and S .
 (b) Find the matrix B representing T relative to the bases S and E .
 (We can view T as a linear mapping from one space into another, each having its own basis.)
 (a) From Problem 6.2, $(a, b) = (-5a + 2b)u_1 + (3a - b)u_2$. Hence,

$$\begin{aligned} T(e_1) &= T(1, 0) = (2, 1) = -8u_1 + 5u_2 \\ T(e_2) &= T(0, 1) = (-3, 4) = 23u_1 - 13u_2 \end{aligned} \quad \text{and so} \quad A = \begin{bmatrix} -8 & 23 \\ 5 & -13 \end{bmatrix}$$

(b) We have

$$\begin{aligned} T(u_1) &= T(1, 3) = (-7, 13) = -7e_1 + 13e_2 \\ T(u_2) &= T(2, 5) = (-11, 22) = -11e_1 + 22e_2 \end{aligned} \quad \text{and so} \quad B = \begin{bmatrix} -7 & -11 \\ 13 & 22 \end{bmatrix}$$

6.35. How are the matrices A and B in Problem 6.34 related?

By Theorem 6.12, the matrices A and B are equivalent to each other; that is, there exist nonsingular matrices P and Q such that $B = Q^{-1}AP$, where P is the change-of-basis matrix from S to E , and Q is the change-of-basis matrix from E to S . Thus,

$$\begin{aligned} P &= \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}, & Q &= \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}, & Q^{-1} &= \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \\ \text{and} & & Q^{-1}AP &= \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} -8 & -23 \\ 5 & -13 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} = \begin{bmatrix} -7 & -11 \\ 13 & 22 \end{bmatrix} = B \end{aligned}$$

6.36. Prove Theorem 6.14: Let $F: V \rightarrow U$ be linear and, say, $\text{rank}(F) = r$. Then there exist bases V and U such that the matrix representation of F has the following form, where I_r is the r -square identity matrix:

$$A = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

Suppose $\dim V = m$ and $\dim U = n$. Let W be the kernel of F and U' the image of F . We are given that $\text{rank}(F) = r$. Hence, the dimension of the kernel of F is $m - r$. Let $\{w_1, \dots, w_{m-r}\}$ be a basis of the kernel of F and extend this to a basis of V :

$$\{v_1, \dots, v_r, w_1, \dots, w_{m-r}\}$$

Set

$$u_1 = F(v_1), \quad u_2 = F(v_2), \quad \dots, \quad u_r = F(v_r)$$

Then $\{u_1, \dots, u_r\}$ is a basis of U' , the image of F . Extend this to a basis of U , say

$$\{u_1, \dots, u_r, u_{r+1}, \dots, u_n\}$$

Observe that

$$\begin{aligned} F(v_1) &= u_1 = 1u_1 + 0u_2 + \cdots + 0u_r + 0u_{r+1} + \cdots + 0u_n \\ F(v_2) &= u_2 = 0u_1 + 1u_2 + \cdots + 0u_r + 0u_{r+1} + \cdots + 0u_n \\ &\dots\dots\dots \\ F(v_r) &= u_r = 0u_1 + 0u_2 + \cdots + 1u_r + 0u_{r+1} + \cdots + 0u_n \\ F(w_1) &= 0 = 0u_1 + 0u_2 + \cdots + 0u_r + 0u_{r+1} + \cdots + 0u_n \\ &\dots\dots\dots \\ F(w_{m-r}) &= 0 = 0u_1 + 0u_2 + \cdots + 0u_r + 0u_{r+1} + \cdots + 0u_n \end{aligned}$$

Thus, the matrix of F in the above bases has the required form.

SUPPLEMENTARY PROBLEMS

Matrices and Linear Operators

6.37. Let $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be defined by $F(x, y) = (4x + 5y, 2x - y)$.

- Find the matrix A representing F in the usual basis E .
- Find the matrix B representing F in the basis $S = \{u_1, u_2\} = \{(1, 4), (2, 9)\}$.
- Find P such that $B = P^{-1}AP$.
- For $v = (a, b)$, find $[v]_S$ and $[F(v)]_S$. Verify that $[F]_S[v]_S = [F(v)]_S$.

6.38. Let $A: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be defined by the matrix $A = \begin{bmatrix} 5 & -1 \\ 2 & 4 \end{bmatrix}$.

- Find the matrix B representing A relative to the basis $S = \{u_1, u_2\} = \{(1, 3), (2, 8)\}$. (Recall that A represents the mapping A relative to the usual basis E .)
- For $v = (a, b)$, find $[v]_S$ and $[A(v)]_S$.

6.39. For each linear transformation L on $\mathbf{R}^2$, find the matrix A representing L (relative to the usual basis of $\mathbf{R}^2$):

- L is the rotation in $\mathbf{R}^2$ counterclockwise by 45° .
- L is the reflection in $\mathbf{R}^2$ about the line $y = x$.
- L is defined by $L(1, 0) = (3, 5)$ and $L(0, 1) = (7, -2)$.
- L is defined by $L(1, 1) = (3, 7)$ and $L(1, 2) = (5, -4)$.

6.40. Find the matrix representing each linear transformation T on $\mathbf{R}^3$ relative to the usual basis of $\mathbf{R}^3$:

- $T(x, y, z) = (x, y, 0)$.
- $T(x, y, z) = (z, y + z, x + y + z)$.
- $T(x, y, z) = (2x - 7y - 4z, 3x + y + 4z, 6x - 8y + z)$.

6.41. Repeat Problem 6.40 using the basis $S = \{u_1, u_2, u_3\} = \{(1, 1, 0), (1, 2, 3), (1, 3, 5)\}$.

6.42. Let L be the linear transformation on $\mathbf{R}^3$ defined by

$$L(1, 0, 0) = (1, 1, 1), \quad L(0, 1, 0) = (1, 3, 5), \quad L(0, 0, 1) = (2, 2, 2)$$

- Find the matrix A representing L relative to the usual basis of $\mathbf{R}^3$.
- Find the matrix B representing L relative to the basis S in Problem 6.41.

6.43. Let $\mathbf{D}$ denote the differential operator; that is, $\mathbf{D}(f(t)) = df/dt$. Each of the following sets is a basis of a vector space V of functions. Find the matrix representing $\mathbf{D}$ in each basis:

- $\{e^t, e^{2t}, te^{2t}\}$.
- $\{1, t, \sin 3t, \cos 3t\}$.
- $\{e^{5t}, te^{5t}, t^2e^{5t}\}$.

6.44. Let $\mathbf{D}$ denote the differential operator on the vector space V of functions with basis $S = \{\sin \theta, \cos \theta\}$.

- (a) Find the matrix $A = [\mathbf{D}]_S$. (b) Use A to show that $\mathbf{D}$ is a zero of $f(t) = t^2 + 1$.

6.45. Let V be the vector space of 2×2 matrices. Consider the following matrix M and usual basis E of V :

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{and} \quad E = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

Find the matrix representing each of the following linear operators T on V relative to E :

- (a) $T(A) = MA$. (b) $T(A) = AM$. (c) $T(A) = MA - AM$.

6.46. Let $\mathbf{1}_V$ and $\mathbf{0}_V$ denote the identity and zero operators, respectively, on a vector space V . Show that, for any basis S of V , (a) $[\mathbf{1}_V]_S = I$, the identity matrix. (b) $[\mathbf{0}_V]_S = 0$, the zero matrix.

Change of Basis

6.47. Find the change-of-basis matrix P from the usual basis E of $\mathbf{R}^2$ to a basis S , the change-of-basis matrix Q from S back to E , and the coordinates of $v = (a, b)$ relative to S , for the following bases S :

- (a) $S = \{(1, 2), (3, 5)\}$. (c) $S = \{(2, 5), (3, 7)\}$.
 (b) $S = \{(1, -3), (3, -8)\}$. (d) $S = \{(2, 3), (4, 5)\}$.

6.48. Consider the bases $S = \{(1, 2), (2, 3)\}$ and $S' = \{(1, 3), (1, 4)\}$ of $\mathbf{R}^2$. Find the change-of-basis matrix:

- (a) P from S to S' . (b) Q from S' back to S .

6.49. Suppose that the x -axis and y -axis in the plane $\mathbf{R}^2$ are rotated counterclockwise 30° to yield new x' -axis and y' -axis for the plane. Find

- (a) The unit vectors in the direction of the new x' -axis and y' -axis.
 (b) The change-of-basis matrix P for the new coordinate system.
 (c) The new coordinates of the points $A(1, 3)$, $B(2, -5)$, $C(a, b)$.

6.50. Find the change-of-basis matrix P from the usual basis E of $\mathbf{R}^3$ to a basis S , the change-of-basis matrix Q from S back to E , and the coordinates of $v = (a, b, c)$ relative to S , where S consists of the vectors:

- (a) $u_1 = (1, 1, 0)$, $u_2 = (0, 1, 2)$, $u_3 = (0, 1, 1)$.
 (b) $u_1 = (1, 0, 1)$, $u_2 = (1, 1, 2)$, $u_3 = (1, 2, 4)$.
 (c) $u_1 = (1, 2, 1)$, $u_2 = (1, 3, 4)$, $u_3 = (2, 5, 6)$.

6.51. Suppose S_1, S_2, S_3 are bases of V . Let P and Q be the change-of-basis matrices, respectively, from S_1 to S_2 and from S_2 to S_3 . Prove that PQ is the change-of-basis matrix from S_1 to S_3 .

Linear Operators and Change of Basis

6.52. Consider the linear operator F on $\mathbf{R}^2$ defined by $F(x, y) = (5x + y, 3x - 2y)$ and the following bases of $\mathbf{R}^2$:

$$S = \{(1, 2), (2, 3)\} \quad \text{and} \quad S' = \{(1, 3), (1, 4)\}$$

- (a) Find the matrix A representing F relative to the basis S .
 (b) Find the matrix B representing F relative to the basis S' .
 (c) Find the change-of-basis matrix P from S to S' .
 (d) How are A and B related?

6.53. Let $A: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be defined by the matrix $A = \begin{bmatrix} 1 & -1 \\ 3 & 2 \end{bmatrix}$. Find the matrix B that represents the linear operator A relative to each of the following bases: (a) $S = \{(1, 3)^T, (2, 5)^T\}$. (b) $S = \{(1, 3)^T, (2, 4)^T\}$.

6.54. Let $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be defined by $F(x, y) = (x - 3y, 2x - 4y)$. Find the matrix A that represents F relative to each of the following bases: (a) $S = \{(2, 5), (3, 7)\}$. (b) $S = \{(2, 3), (4, 5)\}$.

6.55. Let $A: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be defined by the matrix $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 7 & 4 \\ 1 & 4 & 3 \end{bmatrix}$. Find the matrix B that represents the linear operator A relative to the basis $S = \{(1, 1, 1)^T, (0, 1, 1)^T, (1, 2, 3)^T\}$.

Similarity of Matrices

6.56. Let $A = \begin{bmatrix} 1 & 1 \\ 2 & -3 \end{bmatrix}$ and $P = \begin{bmatrix} 1 & -2 \\ 3 & -5 \end{bmatrix}$.

(a) Find $B = P^{-1}AP$. (b) Verify that $\text{tr}(B) = \text{tr}(A)$. (c) Verify that $\det(B) = \det(A)$.

6.57. Find the trace and determinant of each of the following linear maps on $\mathbf{R}^2$:

(a) $F(x, y) = (2x - 3y, 5x + 4y)$. (b) $G(x, y) = (ax + by, cx + dy)$.

6.58. Find the trace and determinant of each of the following linear maps on $\mathbf{R}^3$:

(a) $F(x, y, z) = (x + 3y, 3x - 2z, x - 4y - 3z)$.

(b) $G(x, y, z) = (y + 3z, 2x - 4z, 5x + 7y)$.

6.59. Suppose $S = \{u_1, u_2\}$ is a basis of V , and $T: V \rightarrow V$ is defined by $T(u_1) = 3u_1 - 2u_2$ and $T(u_2) = u_1 + 4u_2$. Suppose $S' = \{w_1, w_2\}$ is a basis of V for which $w_1 = u_1 + u_2$ and $w_2 = 2u_1 + 3u_2$.

(a) Find the matrices A and B representing T relative to the bases S and S' , respectively.

(b) Find the matrix P such that $B = P^{-1}AP$.

6.60. Let A be a 2×2 matrix such that only A is similar to itself. Show that A is a scalar matrix, that is, that $A = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$.

6.61. Show that all matrices similar to an invertible matrix are invertible. More generally, show that similar matrices have the same rank.

Matrix Representation of General Linear Mappings

6.62. Find the matrix representation of each of the following linear maps relative to the usual basis for $\mathbf{R}^n$:

(a) $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ defined by $F(x, y, z) = (2x - 4y + 9z, 5x + 3y - 2z)$.

(b) $F: \mathbf{R}^2 \rightarrow \mathbf{R}^4$ defined by $F(x, y) = (3x + 4y, 5x - 2y, x + 7y, 4x)$.

(c) $F: \mathbf{R}^4 \rightarrow \mathbf{R}$ defined by $F(x_1, x_2, x_3, x_4) = 2x_1 + x_2 - 7x_3 - x_4$.

6.63. Let $G: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ be defined by $G(x, y, z) = (2x + 3y - z, 4x - y + 2z)$.

(a) Find the matrix A representing G relative to the bases

$S = \{(1, 1, 0), (1, 2, 3), (1, 3, 5)\}$ and $S' = \{(1, 2), (2, 3)\}$

(b) For any $v = (a, b, c)$ in $\mathbf{R}^3$, find $[v]_S$ and $[G(v)]_{S'}$. (c) Verify that $A[v]_S = [G(v)]_{S'}$.

6.64. Let $H: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be defined by $H(x, y) = (2x + 7y, x - 3y)$ and consider the following bases of $\mathbf{R}^2$:

$S = \{(1, 1), (1, 2)\}$ and $S' = \{(1, 4), (1, 5)\}$

(a) Find the matrix A representing H relative to the bases S and S' .

(b) Find the matrix B representing H relative to the bases S' and S .

- 6.65.** Let $F: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ be defined by $F(x, y, z) = (2x + y - z, 3x - 2y + 4z)$.
- (a) Find the matrix A representing F relative to the bases
 $S = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$ and $S' = (1, 3), (1, 4)\}$
- (b) Verify that, for any $v = (a, b, c)$ in $\mathbf{R}^3$, $A[v]_S = [F(v)]_{S'}$.
- 6.66.** Let S and S' be bases of V , and let $\mathbf{1}_V$ be the identity mapping on V . Show that the matrix A representing $\mathbf{1}_V$ relative to the bases S and S' is the inverse of the change-of-basis matrix P from S to S' ; that is, $A = P^{-1}$.
- 6.67.** Prove (a) Theorem 6.10, (b) Theorem 6.11, (c) Theorem 6.12, (d) Theorem 6.13. [Hint: See the proofs of the analogous Theorems 6.1 (Problem 6.9), 6.2 (Problem 6.10), 6.3 (Problem 6.11), and 6.7 (Problem 6.26).]

Miscellaneous Problems

- 6.68.** Suppose $F: V \rightarrow V$ is linear. A subspace W of V is said to be *invariant* under F if $F(W) \subseteq W$. Suppose W is invariant under F and $\dim W = r$. Show that F has a block triangular matrix representation $M = \begin{bmatrix} A & B \\ 0 & C \end{bmatrix}$ where A is an $r \times r$ submatrix.
- 6.69.** Suppose $V = U + W$, and suppose U and W are each invariant under a linear operator $F: V \rightarrow V$. Also, suppose $\dim U = r$ and $\dim W = s$. Show that F has a block diagonal matrix representation $M = \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}$ where A and B are $r \times r$ and $s \times s$ submatrices.
- 6.70.** Two linear operators F and G on V are said to be *similar* if there exists an invertible linear operator T on V such that $G = T^{-1} \circ F \circ T$. Prove
- (a) F and G are similar if and only if, for any basis S of V , $[F]_S$ and $[G]_S$ are similar matrices.
- (b) If F is diagonalizable (similar to a diagonal matrix), then any similar matrix G is also diagonalizable.

ANSWERS TO SUPPLEMENTARY PROBLEMS

Notation: $M = [R_1; R_2; \dots]$ represents a matrix M with rows $R_1, R_2, \dots$.

- 6.37.** (a) $A = [4, 5; 2, -1]$; (b) $B = [220, 487; -98, -217]$; (c) $P = [1, 2; 4, 9]$;
 (d) $[v]_S = [9a - 2b, -4a + b]^T$ and $[F(v)]_S = [32a + 47b, -14a - 21b]^T$
- 6.38.** (a) $B = [-6, -28; 4, 15]$;
 (b) $[v]_S = [4a - b, -\frac{3}{2}a + \frac{1}{2}b]^T$ and $[A(v)]_S = [18a - 8b, \frac{1}{2}(-13a + 7b)]$
- 6.39.** (a) $[\sqrt{2}, -\sqrt{2}; \sqrt{2}, \sqrt{2}]$; (b) $[0, 1; 1, 0]$; (c) $[3, 7; 5, -2]$;
 (d) $[1, 2; 18, -11]$
- 6.40.** (a) $[1, 0, 0; 0, 1, 0; 0, 0, 0]$; (b) $[0, 0, 1; 0, 1, 1; 1, 1, 1]$;
 (c) $[2, -7, -4; 3, 1, 4; 6, -8, 1]$
- 6.41.** (a) $[1, 3, 5; 0, -5, -10; 0, 3, 6]$; (b) $[0, 1, 2; -1, 2, 3; 1, 0, 0]$;
 (c) $[15, 65, 104; -49, -219, -351; 29, 130, 208]$
- 6.42.** (a) $[1, 1, 2; 1, 3, 2; 1, 5, 2]$; (b) $[0; 2, 14, 22; 0, -5, -8]$
- 6.43.** (a) $[1, 0, 0; 0, 2, 1; 0, 0, 2]$; (b) $[0, 1, 0, 0; 0; 0, 0, 0, -3; 0, 0, 3, 0]$;
 (c) $[5, 1, 0; 0, 5, 2; 0, 0, 5]$

6.44. (a) $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$; (b) $A^2 + I = 0$

6.45. (a) $\begin{bmatrix} a & 0 & b & 0 \\ 0 & a & 0 & b \\ c & 0 & d & 0 \\ 0 & c & 0 & d \end{bmatrix}$;
 (b) $\begin{bmatrix} a & c & 0 & 0 \\ b & d & 0 & 0 \\ 0 & 0 & a & c \\ 0 & 0 & b & d \end{bmatrix}$;
 (c) $\begin{bmatrix} 0 & -c & b & 0 \\ -b & a-d & 0 & b \\ c & 0 & d-a & -c \\ 0 & c & -b & 0 \end{bmatrix}$

6.47. (a) $\begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix}$, $\begin{bmatrix} -5 & 3 \\ 2 & -1 \end{bmatrix}$, $[v] = \begin{bmatrix} -5a + 3b & 2a - b \end{bmatrix}^T$;
 (b) $\begin{bmatrix} 1 & 3 \\ -3 & -8 \end{bmatrix}$, $\begin{bmatrix} -8 & -3 \\ 3 & 1 \end{bmatrix}$, $[v] = \begin{bmatrix} -8a - 3b & 3a + b \end{bmatrix}^T$;
 (c) $\begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}$, $\begin{bmatrix} -7 & 3 \\ 5 & -2 \end{bmatrix}$, $[v] = \begin{bmatrix} -7a + 3b & 5a - 2b \end{bmatrix}^T$;
 (d) $\begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix}$, $\begin{bmatrix} -\frac{5}{2} & 2 \\ \frac{3}{2} & -1 \end{bmatrix}$, $[v] = \begin{bmatrix} -\frac{5}{2}a + 2b & \frac{3}{2}a - b \end{bmatrix}^T$

6.48. (a) $P = \begin{bmatrix} 3 & 5 \\ -1 & -2 \end{bmatrix}$; (b) $Q = \begin{bmatrix} 2 & 5 \\ -1 & -3 \end{bmatrix}$

6.49. Here $K = \sqrt{3}$.

(a) $\frac{1}{2}(K, 1)$, $\frac{1}{2}(-1, K)$;
 (b) $P = \frac{1}{2}\begin{bmatrix} K & -1 \\ 1 & K \end{bmatrix}$;
 (c) $\frac{1}{2}[K + 3, 3K - 1]^T$, $\frac{1}{2}[2K - 5, -5K - 2]^T$, $\frac{1}{2}[aK + b, bK - a]^T$

6.50. P is the matrix whose columns are u_1, u_2, u_3 , $Q = P^{-1}$, $[v] = Q[a, b, c]^T$.

(a) $Q = \begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 1 \\ -2 & 2 & -1 \end{bmatrix}$, $[v] = \begin{bmatrix} a \\ a - b + c \\ -2a + 2b - c \end{bmatrix}^T$;
 (b) $Q = \begin{bmatrix} 0 & -2 & 1 \\ 2 & 3 & -2 \\ -1 & -1 & 1 \end{bmatrix}$, $[v] = \begin{bmatrix} -2b + c \\ 2a + 3b - 2c \\ -a - b + c \end{bmatrix}^T$;
 (c) $Q = \begin{bmatrix} -2 & 2 & -1 \\ -7 & 4 & -1 \\ 5 & -3 & 1 \end{bmatrix}$, $[v] = \begin{bmatrix} -2a + 2b - c \\ -7a + 4b - c \\ 5a - 3b + c \end{bmatrix}^T$

6.52. (a) $\begin{bmatrix} -23 & -39 \\ 15 & 26 \end{bmatrix}$; (b) $\begin{bmatrix} 35 & 41 \\ -27 & -32 \end{bmatrix}$; (c) $\begin{bmatrix} 3 & 5 \\ -1 & -2 \end{bmatrix}$; (d) $B = P^{-1}AP$

6.53. (a) $\begin{bmatrix} 28 & 47 \\ -15 & -25 \end{bmatrix}$; (b) $\begin{bmatrix} 13 & 18 \\ -\frac{15}{2} & -10 \end{bmatrix}$

6.54. (a) $\begin{bmatrix} 43 & 60 \\ -33 & -46 \end{bmatrix}$; (b) $\frac{1}{2}\begin{bmatrix} 3 & 7 \\ -5 & -9 \end{bmatrix}$

6.55. $\begin{bmatrix} 10 & 8 & 20 \\ 13 & 11 & 28 \\ -5 & -4 & -10 \end{bmatrix}$

6.56. (a) $\begin{bmatrix} -34 & 57 \\ -19 & 32 \end{bmatrix}$; (b) $\text{tr}(B) = \text{tr}(A) = -2$; (c) $\det(B) = \det(A) = -5$

6.57. (a) $\text{tr}(F) = 6$, $\det(F) = 23$; (b) $\text{tr}(G) = a + d$, $\det(G) = ad - bc$

6.58. (a) $\text{tr}(F) = -2$, $\det(F) = 13$; (b) $\text{tr}(G) = 0$, $\det(G) = 22$

6.59. (a) $A = \begin{bmatrix} 3 & 1 \\ -2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 8 & 11 \\ -2 & -1 \end{bmatrix}$; (b) $P = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$

6.62. (a) $\begin{bmatrix} 2 & -4 & 9 \\ 5 & 3 & -2 \end{bmatrix}$; (b) $\begin{bmatrix} 3 & 5 & 1 & 4 \\ 4 & -2 & 7 & 0 \end{bmatrix}$; (c) $\begin{bmatrix} 2 & 1 & -7 & -1 \end{bmatrix}$

6.63. (a) $\begin{bmatrix} -9 & 1 & 4 \\ 7 & 2 & 1 \end{bmatrix}$; (b) $[v]_S = \begin{bmatrix} -a + 2b - c \\ 5a - 5b + 2c \\ -3a + 3b - c \end{bmatrix}^T$, and $[G(v)]_{S'} = \begin{bmatrix} 2a - 11b + 7c \\ 7b - 4c \end{bmatrix}^T$

6.64. (a) $A = \begin{bmatrix} 47 & 85 \\ -38 & -69 \end{bmatrix}$; (b) $B = \begin{bmatrix} 71 & 88 \\ -41 & -51 \end{bmatrix}$

6.65. $A = \begin{bmatrix} 3 & 11 & 5 \\ -1 & -8 & -3 \end{bmatrix}$